

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifica-
8 tion demonstrating how cognitive architectures can emerge as attrac-
9 tor states from established physical primitives, without introducing
10 new fundamental physics. Starting from the finiteness of a localized
11 system’s phase space, we construct a framework for predictive process-
12 ing; the spacetime symmetries are the setting rather than a premise,
13 and no theorem consumes them. Spontaneous symmetry breaking gener-
14 ates necessary topological boundaries, and finite phase space sub-
15 jects these boundaries to Landauer’s thermodynamic erasure. Agency

16 arises when systems evolve along paths of least action, minimizing en-
 17 tropy production via internal structural resonance with external per-
 18 turbations. Scaling this dynamic to macroscopic levels aligns with
 19 emerging paradigms in neuroscience—specifically classical electrody-
 20 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
 21 tion, culminating in a topological fixed point: a unified “Self.” We
 22 state the claim narrowly, because the subject matter invites a wider
 23 reading than the work supports: no new physics is derived here. Every
 24 physical result the chain invokes is established—Picard–Lindelöf, Kib-
 25 ble’s classification of topological defects, Landauer’s erasure bound,
 26 Gibbs’ inequality, Still’s bound on dissipation by nonpredictive in-
 27 formation, Barbalat’s lemma, Sakaguchi’s threshold, Banach’s fixed
 28 point. What is new is that they are stated in one formal language,
 29 their consistency mechanically checked, and the points where the chain
 30 relies on stipulation rather than derivation located exactly. The chain
 31 itself composes to a single theorem, whose eight unproved steps are its
 32 arguments rather than sentences about it, and which are jointly satis-
 33 fiable rather than jointly empty. The development declares no axioms
 34 and every structure carrying physical content is inhabited by a witness;
 35 the synchronization step is proved from a dynamics for initial data in-
 36 side an explicit basin and assumed outside it, and the identification of
 37 a global section of the probability sheaf with the unity of experience
 38 is a stipulation, marked as such throughout rather than absorbed into
 39 the derivation. The step from a coarse-grained coupling to the brain’s

40 endogenous electromagnetic field is likewise an empirical commitment
41 and not a deduction; we separate it from the coarse-graining theorem
42 that precedes it, weaken it to a modulatory-but-integrative role, and
43 give four conditions that would falsify it. The framework is nonethe-
44 less falsifiable in one place that is its own and no rival's: recovery
45 from sudden awakening should follow a delayed sigmoid rather than
46 an exponential, because the variable gating it is astrocytic volume on
47 a timescale of seconds to minutes rather than cortical phase realign-
48 ment on a timescale of milliseconds. As a corollary, we show why
49 rigid von Neumann architectures (e.g., GPUs) are thermodynamically
50 disadvantaged or structurally constrained from reaching these specific
51 continuous thermodynamic attractors.

52 1 Introduction

53 The pursuit of a physical theory of consciousness typically involves search-
54 ing for the biological correlates of mind or treating subjective experience as
55 an abstract computational algorithm. This paper asks a more fundamental
56 question: If we assume only basic physical primitives—continuous symme-
57 tries (the Poincaré group), geometry, and thermodynamics—is the emergence
58 of a unified, perceiving entity a natural physical attractor?

59 We propose a framework where the structural properties of consciousness
60 are derived as physical attractor states. Establishing a deductive chain from
61 basic principles, we demonstrate how continuous symmetries break to form

62 boundaries, how those boundaries dissipate free energy as non-equilibrium
 63 steady states, and how their evolution along paths of least action results
 64 in physical resonance with their environment. This eventually scales into
 65 the macroscopic electromagnetic unity proposed by emerging field theories
 66 of mind. To avoid critiques of “math theatre,” we delineate between literal
 67 physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle,
 68 boundaries from spontaneous symmetry breaking) and theoretical mappings,
 69 utilizing Lean 4 formalization as our mathematical anchor. We adopt a per-
 70 spective aligned with Russellian Monism: objective measurements of macro-
 71 scopic electromagnetic fields constitute the third-person “outside,” while the
 72 system’s internal physical deformation to reach structural resonance consti-
 73 tutes the first-person intrinsic experience.

74 **What is new here, and what is not**

75 A framework paper about consciousness invites a particular suspicion, and it
 76 is better answered at the start than at the end. **No new physics is derived**
 77 **here.** Every physical result the chain invokes is established, and most of
 78 it is old: Picard–Lindelöf for the well-posedness of the coupled dynamics
 79 (Coddington & Levinson, 1955); Kibble’s classification of topological defects
 80 by the homotopy of the vacuum manifold (Kibble, 1976); Landauer’s bound
 81 on the heat cost of erasure (Landauer, 1961); Gibbs’ inequality for the relative
 82 entropy (Cover & Thomas, 2006); Barbalat’s lemma for the asymptotics of a
 83 dissipating trajectory (Barbalat, 1959); the threshold $K_c = 2D$ for the noisy

84 mean-field Kuramoto model (Sakaguchi, 1988); the bound of Still, Sivak,
 85 Bell and Crooks relating dissipation to nonpredictive information (Still et
 86 al., 2012); the identity of Kawai, Parrondo and Van den Broeck equating
 87 dissipated work with the relative entropy between forward and time-reversed
 88 ensembles (Kawai et al., 2007); and Banach’s fixed-point theorem (Banach,
 89 1922). A physicist reading this paper will recognize all of it and would
 90 establish the point in three paragraphs if we did not. Conceding it costs
 91 nothing.

92 What is new is of a different kind, and it is a smaller claim than a theory
 93 of consciousness usually makes. These results have been stated in one formal
 94 language, their mutual consistency has been mechanically checked, and the
 95 places where the chain relies on stipulation rather than derivation have been
 96 located exactly. The value of a machine-checked chain is not that any link
 97 is novel; it is that the claim being made *about* the chain is itself checkable,
 98 and we state it in the form a reader can check rather than in the form that
 99 asks to be believed. Four things. The development declares **no axioms**:
 100 **#print axioms** on any result reports only **propext**, **Classical.choice** and
 101 **Quot.sound**. Every physical postulate is a field of a class that each model
 102 discharges for itself rather than a global assertion, and every such class is
 103 inhabited by an explicit witness, so no theorem is vacuous for want of a model.
 104 The chain composes to a *single* theorem — **chain**, in **Chain.lean** — which
 105 derives the Self from the finiteness of the phase space and takes each unproved
 106 step as one of **eight named hypotheses**: four formalization gaps, two

107 modelling assumptions and two physical commitments, each an implication
108 between two links rather than an opaque symbol, so that deleting one fails
109 the build instead of softening a sentence. And the eight hold *simultaneously*
110 on one substrate (`chain_nonvacuous`), which rules out the failure in which
111 a composition proves its conclusion only because its premises cannot all be
112 met at once. The seams — where physics stops and interpretation begins —
113 are thereby fixed in the source rather than negotiable in the prose, and their
114 number is a number rather than an impression.

115 Composing the chain also refuted two of its own arrows, and Figure 1
116 draws the corrected picture rather than repairing it quietly. The capacity
117 bound of the first link has no hypotheses at all, so an implication from it to
118 the existence of a boundary is equivalent to assuming the boundary outright:
119 those two links are independent starting points and not consecutive steps.
120 What finiteness does imply directly is the heat cost, skipping the boundary
121 entirely. Neither of the two arrows that are theorems is one the chain’s
122 usual narration draws. Section 2.1 reports what the check cost: three of five
123 original axioms each proved **False**, from one shared root cause, and none of
124 the three had been caught by reading.

125 The framework is not thereby unfalsifiable, and one of its predictions is
126 its own rather than a rival’s. Section 11 argues that recovery from sudden
127 awakening should follow a *delayed sigmoid* rather than an exponential, be-
128 cause the variable gating recovery is astrocytic volume and cerebrospinal-fluid
129 clearance, on a timescale of seconds to minutes, while the cortical rhythms

130 whose realignment is usually taken to constitute the recovery move in mil-
131 liseconds. That is a claim about functional form; it is fittable against data of
132 a kind already collected; and a purely electrical account of the same recovery
133 predicts a different form.

134 **2 Formalization in Lean 4**

135 A core contribution of this work is the verification of our deductive chain
136 using the Lean 4 interactive theorem prover. This approach builds on recent
137 efforts to formalize consciousness and brain semantics, such as the 'OnlyOne'
138 system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué,
139 2026). It is necessary to explicitly distinguish which claims have been math-
140 ematically derived as theorems within the proof assistant, and which remain
141 irreducible physical postulates. Figure 1 shows the chain with the status of
142 every node *and of every arrow* marked, and Table 1 states in seven rows
143 what each link assumes. The complete claim-by-claim map to Lean identi-
144 fiers, which runs to several pages, is Table S1 in the supplement; the main
145 text does not depend on reading it.

146 **2.1 Soundness of the formalization**

147 Formalization is only as trustworthy as the axioms it rests on, and an axiom
148 that is inconsistent silently makes every theorem downstream of it vacuous
149 while leaving the build green. We report a case of exactly this, found by

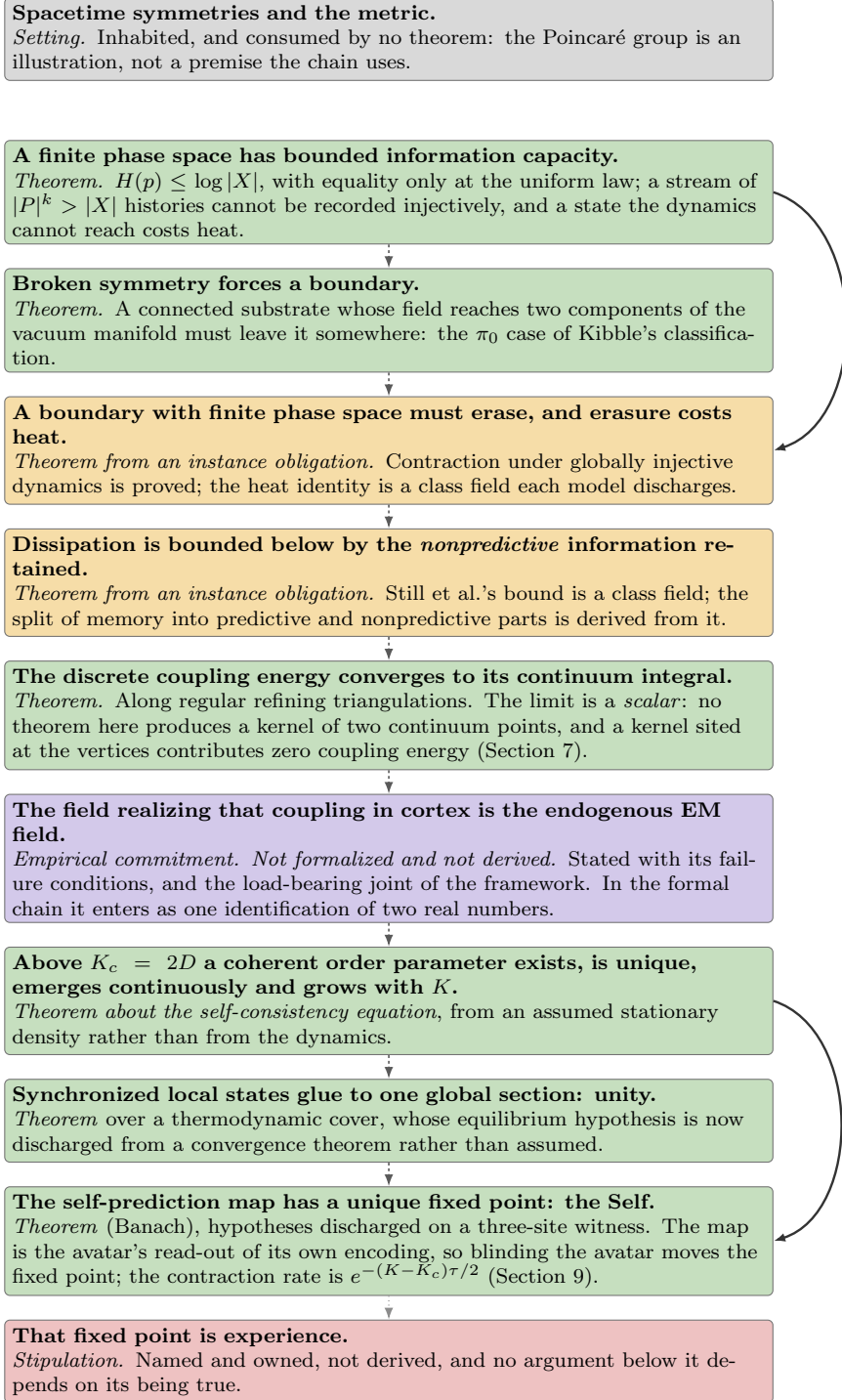


Figure 1: The deductive chain, with⁸ the status of each node and of each arrow. *Nodes.* ■ theorem; ■ theorem resting on a class field that each model must discharge; ■ empirical commitment, neither formalized nor derived; ■ stipulation; ■ setting, consumed by no theorem. *Arrows.* A **solid** arrow is a Lean theorem. A **dashed** arrow is a named hypothesis of chain (Chain.lean): an implication between the two node statements which the development does not prove and which is therefore an argument the theorem

Claim	Status	What is assumed
A finite phase space has bounded information capacity	Theorem	Finiteness of the state space, and nothing else — no symmetry group, no metric. $H \leq \log X $, with equality only at the uniform law. On an infinite state space the step from “unreachable state” to “erasure” fails, and the failure is exhibited rather than assumed away.
Broken symmetry forces a boundary	Theorem	Substrate connected; vacuum manifold disconnected. No hypothesis mentions energy, a potential, or a symmetry.
Erasure costs heat	Theorem + class field	Global system–bath dynamics injective (Liouville). The heat identity <code>heat_eq</code> is discharged per instance, not asserted for all models.
Dissipation is bounded below by nonpredictive information	Theorem + class field	Still et al.’s bound is discharged per instance; the environment evolves by a Markov kernel the system does not act on.
The discrete coupling energy converges to its continuum integral; the field carrying that coupling is the cortical EM field	Theorem; then empirical	Convergence holds along regular refining triangulations, for the energy functional and not the dynamics. The limit is a scalar: no theorem produces a kernel of two continuum points, and a kernel sited at the triangulation’s vertices contributes zero coupling energy on an atomless substrate. The physical identification is a commitment with stated failure conditions (Section 7).
Above $K_c = 2D$ a unique coherent state exists, and synchronized patches glue into a unique global section	Theorem 9	Von Mises stationary density assumed rather than derived from the Fokker–Planck operator; couplings positive and symmetric, so unfrustrated. Gluing needs a cover at the potential minimum whose synchronized patches agree on overlaps; the first is derived from a convergent trajectory, the second is an instance obligation throughout.
The coherent order parameter carries a unique reflexive fixed point:	Theorem	The self-prediction map is the avatar’s read-out of its own encoding, and its Lipschitz rate

auditing our own development, because the failure mode is easy to reproduce and we have not seen it discussed in the applied-formalization literature.

Earlier drafts of this work declared five axioms. Three of them were individually inconsistent: each, on its own, proves `False`. All three shared one root cause — an axiom that constrains a symbol it does not itself bind. Concretely:

- `spontaneous_symmetry_breaking_pointwise_min` took the potential-energy functional as an *implicit* argument occurring only in its hypothesis `PotentialEnergy phi = V v0`. Instantiating that functional with the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$ this proves $1 = 0$.
- `landauer_heat_eq` equated the free class field `heat_dissipation` with a bath-entropy difference *for every instance* of the class. An instance with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.
- `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P||Q)/\Delta t$ for *all* P and Q , while σ is a single real fixed by a class field. Since D_{KL} is unbounded above, a point mass against a uniform distribution refutes it.

The remaining two axioms, `phase_invariant_periodic` and `sync_to_section_eq`, had the same shape — global assertions pinning free fields of `LocalSectionSynchronization` — and are refutable as soon as the probability presheaf admits two distinct

171 global sections. (Both were demoted to class fields at the time; both have
 172 since been removed from the class altogether, for a separate reason given in
 173 Derivation 5.)

174 The correction is uniform, and we state it as a rule we now follow: *a*
 175 *physical postulate that mentions a class field must be a field of that class,*
 176 *never a standalone axiom quantified over all instances.* A postulate should
 177 be an obligation each model discharges, not a global claim about every model.
 178 Recast this way, the development declares no axioms at all. Two of the five
 179 became genuine theorems — pointwise vacuum minimization is now proved
 180 from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs’ inequality was already one
 181 — and the other three became instance obligations.

182 This makes non-vacuity the live question, since an uninhabitable class
 183 makes its theorems vacuous just as surely as an inconsistent axiom did.
 184 **Examples.lean** therefore constructs witnesses: a one-bit erasure system with
 185 a genuine bath that discharges **heat_eq** and reproduces $\Delta Q = k_B T \ln 2$;
 186 a perfectly resonant system discharging **kl_bound**; and a scalar field on
 187 a one-point spacetime discharging **ActionPrinciples**. Derivation 5’s glu-
 188 ing structures are witnessed too, which was until recently the largest open
 189 gap: **Examples.lean** builds a **ThermodynamicCover** over a three-site sub-
 190 strate carrying the discrete topology and its Borel σ -algebra, covered by two
 191 *overlapping* open patches whose intersection is a single shared site. The
 192 local sections are germ families of a phase-indexed finite measure $\mu_\theta =$
 193 $(1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed into the sheafification of the probability presheaf; the

194 family is provably non-constant in θ (`phaseMeasure_not_const`), so the pe-
 195 riodicity obligation of the constructor `LocalSectionSynchronization.of-`
 196 `InvariantMeasure` is discharged by an argument about `cos` rather than satis-
 197 fied trivially, and `thermodynamic_equilibrium` is *derived* from `phase_locked_minimizes_potential`
 198 rather than assumed. `global_section_from_thermodynamics` then fires on
 199 this instance and glues the two patch-local sections into a unique global one.
 200 Two further covers answer the follow-up question of how non-degenerate such
 201 a witness can be made. `Examples.lean` §13 has a negative half worth record-
 202 ing: the phase field can be made non-constant as a function, but not as a
 203 physical state, because `thermodynamic_equilibrium` forces every instance
 204 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 205 round, from two patches with independently chosen mass profiles that meet
 206 only on the site they share, and the section the gluing returns is neither of
 207 them. That is the same kind of result as `contracting_implies_const` in
 208 Derivation 6 — a proof that the witness is as strong as the class permits,
 209 rather than a witness that could be improved by trying harder.

210 What the witnesses establish is that the classes are inhabitable, so the
 211 theorems about them are not vacuous. What they do not establish is that
 212 a cortex is such an instance. In the cover above the phase field is already
 213 constant — every patch locked at $\theta = 0$ — which is precisely what makes
 214 the equilibrium obligation dischargeable; getting a physical system into that
 215 configuration is the content of the informal dynamical argument, not of the
 216 Lean development.

217 The same audit turned up a defect of a different kind, worth record-
 218 ing because it is not an axiom problem and would not have been caught
 219 by `#print axioms. mesh_refinement_convergence` — the claim that dis-
 220 crete edge weights converge to the continuous energy as the mesh is refined
 221 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
 222 not merely unproved but false. Nothing required the edge regions to cover
 223 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
 224 all empty satisfied the mesh hypothesis while contributing nothing to the
 225 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
 226 quantified over a triangulation and then resolved the vertex type by instance
 227 search rather than through the bound variable, so the variable was never
 228 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
 229 gence is now stated along a *sequence* of triangulations of a named region
 230 S , each required to be regular — edge regions measurable, without self-
 231 loops, disjoint across distinct edges, covering S , and anchored to the sim-
 232 plicial data in the sense that a vertex lies in the closure of every region
 233 of an edge at it. Closure rather than membership is forced: cells are dis-
 234 joint, so a shared vertex belongs to at most one of them. The manuscript’s
 235 discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
 236 and is therefore not symmetric, is shown to equal exactly the midpoint rule
 237 $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
 238 tion, reading the double sum as a Riemann sum would be a category error.
 239 At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of

240 the integrand, and convergence follows for uniformly continuous integrands
 241 as the mesh shrinks. The regularity conditions pull against each other, so
 242 `Examples.lean` §6 discharges them with the uniform partition of $[0, 1)$ into
 243 half-open cells and checks that the approximation genuinely moves $-\frac{1}{2}$ at
 244 $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
 245 converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
 246 the accompanying simulation measures; that would need a midpoint error
 247 term with a second derivative, which we have not developed.

248 A third defect was one of bookkeeping across files, and it is the kind that
 249 survives any number of green builds because each half is individually cor-
 250 rect. The development carried *two* Kuramoto potentials. Lyapunov descent
 251 along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
 252 $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum
 253 — phase-locked in both directions — and everything downstream of it, in-
 254 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 255 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 256 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 257 — walk θ out along ω and the bounded cosine sum cannot compensate — so
 258 it has no minimum for phase-locking to attain, and the sentence “the phase-
 259 locked state minimizes the Lyapunov potential” was, read literally, false of
 260 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 261 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical
 262 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-

frequency system — which is exactly the system Derivation 5 already worked
 with — the two potentials coincide there, and the descent result transports
 onto the coupling potential whose minimizers are the phase-locked states.
 Phase differences, phase-locking and the magnitude of the order parameter
 are all proved invariant under the change of frame, so nothing observable
 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 two oscillators with a common frequency, discharging every hypothesis. The
 restriction is real and we state it plainly: the reduction is exact only for
 identical frequencies. A genuine spread of natural frequencies leaves residual
 detunings in the reduced system, phase-locking then requires coupling above
 a critical K_c , and that K_c — Kuramoto’s threshold for a spread of frequen-
 cies, not the noise threshold $2D$ of Section 10 — is nowhere in our Lean
 development. It is studied numerically, in Figure 5, as Table S1 records.

A fourth defect was dimensional, and it concerned the one row of Table S1
 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 against an unnormalized measure carries the substrate’s mass twice over,
 so a constant kernel K on a substrate of total mass m contributed Km^2 .
 The comparison therefore depended on how large the substrate was rather
 than on how strongly it was coupled, and any kernel whatever could be
 pushed above threshold by inflating m . `Examples.lean` §9 exhibits the fail-
 ure concretely: on a two-site substrate carrying counting measure, a sys-

286 tem coupled at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two
 287 *below* threshold — had an unnormalized double integral of $2 > 1$ and so
 288 satisfied the old predicate. The fix is to require the substrate’s measure to
 289 be a probability measure, which is the normalization in which the mean-
 290 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 291 then gives that a constant kernel has mean-field strength exactly K , and
 292 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 293 We flag what this does and does not do. It makes the predicate say something
 294 whose truth depends only on the physics, and §9 witnesses it on both sides of
 295 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 296 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 297 at $K = 2D$, which is a separate question from whether the predicate is di-
 298 mensionally sound. That bifurcation is proved below, in both directions,
 299 from an assumed stationary density; deriving the density itself needs the
 300 Fokker–Planck operator and is not done.

301 We draw no grand moral from this beyond the obvious one. The value
 302 of a proof assistant here was not that it certified our argument, but that it
 303 made a specific, checkable claim about what had been assumed — and that
 304 claim turned out to be false in a way no amount of reading the prose would
 305 have revealed.

306 **3 Axiom 1: Symmetries as Setting, and the** 307 **Capacity of a Finite Phase Space**

308 Our starting premise relies on fundamental geometry. In our Lean formaliza-
309 tion (`Phase1_Primitives.lean`), continuous spacetime symmetries (e.g., the
310 Poincaré group) act upon field states, and the invariance of the system’s total
311 energy under that action is recorded by the class `SymmetryInvariantAction`.
312 Physically, it is such an invariance that Noether’s theorem converts into the
313 conservation of energy and momentum. The scope of the *formal* claim is nar-
314 rower and we state it plainly: the invariance is a class field, i.e. an assumption
315 a model must discharge, and no conserved quantity is ever constructed from
316 it. There is no Noether theorem in our Lean, and nothing downstream of this
317 section depends on one. The class is now inhabited — `Examples.lean` §11
318 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
319 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
320 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
321 changed by the symmetry, so an energy-minimising field and its image have
322 equal energy and differ. That is symmetry breaking exhibited rather than
323 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
324 no one-parameter family to differentiate along in the first place. Simultane-
325 ously, discrete permutation symmetries dictate the statistical mechanics of
326 any physical system, defining a strictly finite volume of phase space.¹

¹While the Poincaré group and discrete permutations serve as accessible physical ex-
amples, the underlying Lean formalization does not hardcode specific physics. It defines

327 Because phase space is finite, any localized physical system possesses a
 328 mathematical limit on the amount of information (distinguishable states) it
 329 can embody at any given moment. That sentence used to be this section’s
 330 entire contribution to the chain, and no theorem carried it. It is now car-
 331 ried by three, in `Phase1.PhaseSpaceCapacity.lean`, whose only hypothesis
 332 about the system is that its state space is finite: no symmetry group, no
 333 metric, no action functional appears anywhere in them.

334 The first is the capacity bound itself. A probability distribution p on a
 335 finite state space X satisfies $H(p) = -\sum_x p(x) \log p(x) \leq \log |X|$ (`shannon_`
 336 `entropy_le_log_card`), with equality if and only if p is uniform (`shannon_`
 337 `entropy_eq_log_card_iff`). The “only if” half is the part worth stating: it
 338 says $\log |X|$ is the exact capacity rather than a convenient over-estimate, so
 339 a system holding any distribution other than the flat one over its states holds
 340 strictly less than its size suggests. The proof is Gibbs’ inequality against the
 341 uniform law — $\log x \leq x - 1$ applied at $x = 1/(|X|p(x))$, with the $p(x) = 0$
 342 case carrying the slack that makes the equality condition sharp — and it is
 343 elementary. What was missing here was the statement, not the difficulty.

344 The second says the limit binds. Let the boundary absorb a stream of
 the system at a more primitive mathematical level, relying entirely on generic Pseudo-
 Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechan-
 ics). What the formalization supplies is the vocabulary in which symmetries, conserved
 quantities and phase-space constraints can be *stated*; it does not derive them. All three
 of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its in-
 variance in §11, the real line with the Euclidean metric in §12 — but the instances are
 witnesses of inhabitability, not of physics: no theorem in the development consumes a
 symmetry group or a metric, and that half of the premise is setting rather than content.
 The other half of the premise no longer is, and the paragraphs below are what changed.

345 perturbations drawn from an alphabet P ; a stream of length k has $|P|^k$
 346 possible histories. Once $|X| < |P|^k$, the map from histories to final states
 347 cannot be injective (`absorb_not_injective`), so two distinct histories leave
 348 the system in the same state, and the entropy of the source strictly exceeds
 349 anything the system can hold (`source_entropy_exceeds_capacity`). The
 350 update map is arbitrary and is allowed to be reversible at every individual
 351 step; the collision is forced by counting alone, which is the sense in which the
 352 premise, and not the model of the dynamics, is doing the work.

353 The third is what Derivation 2 consumes, and it is the one place where
 354 finiteness does something no other hypothesis could. On a finite state space,
 355 a map that fails to reach every state is not injective (`is_erasure_of_not_`
 356 `surjective`) — which is exactly the erasure condition Landauer’s bound re-
 357 quires — so any update of a finite phase space leaving some state unreachable
 358 dissipates strictly positive heat (`finite_phase_space_dissipates`). On an
 359 infinite phase space the step simply fails: the successor map on $\mathbb{N}$ misses 0 and
 360 is injective. `Examples.lean` §19 records that failure next to the witnesses, so
 361 the finiteness hypothesis is fenced rather than assumed to be load-bearing,
 362 and runs all three results on a single bit: the fair coin attains $\log 2$, a $3/4$ -
 363 $1/4$ bit falls strictly short, two binary perturbations already outrun a one-bit
 364 memory, and §1’s one-bit thermodynamic system pays heat for a state it
 365 cannot reach.

366 None of this derives the finiteness of phase space. Finiteness is assumed,
 367 as it is in any statistical-mechanical treatment, and the Poincaré group re-

368 mains an illustration in prose. What has been removed is the gap between
369 assuming the premise and using it.

370 4 Derivation 1: Symmetry Breaking and the 371 Inevitability of Boundaries

372 A perfectly symmetric universe is the uniform vacuum; no distinct entities
373 exist. For any dynamic process to occur, the continuous symmetries of the
374 background field must undergo spontaneous symmetry breaking.

375 When a field drops to a lower, asymmetric energy state, topology dic-
376 tates the natural creation of defects (such as domain walls or monopoles).
377 This constraint necessitates the creation of a physical boundary—a localized
378 topological distinction between an “inside” and an “outside.” While such
379 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
380 entity, it is not sufficient for agency. The boundary is relational, defined by
381 the twisted geometry of the field surrounding it, and must be coupled with
382 non-equilibrium thermodynamics to exhibit active behaviors.

383 That first sentence was, until recently, the one claim of this section that
384 the formalization did not make. What Lean established was the energy half
385 — an energy-minimizing field sits in the vacuum manifold almost everywhere
386 (`pointwise_vacuum_of_global_min`) — together with the observation that
387 a boundary defect *obstructs* extension over a contractible interior. Neither
388 produces a defect, so “topology dictates the natural creation” was asserted

389 rather than proved. It is now the theorem `exists_notMem_of_connected-`
 390 `ComponentIn`: if the substrate is connected and a continuous field takes val-
 391 ues in two *different connected components* of the vacuum manifold M , then
 392 somewhere the field is not in M at all. That excluded region is the wall.
 393 This is the π_0 case of Kibble’s classification of topological defects by the ho-
 394 motopy of the vacuum manifold (Kibble, 1976), and the proof is elementary
 395 in exactly the way it is elementary in the physics literature: the continuous
 396 image of a connected space is connected, a connected subset of M lies in-
 397 side one component of M , and the field’s range would be such a subset if it
 398 never left M . No hypothesis mentions energy, a potential, or a symmetry;
 399 connectedness of the substrate and disconnectedness of the vacuum are the
 400 whole of it.

401 We formalize the π_0 case and stop there. The higher cases — $\pi_1 \neq 0$
 402 forcing vortex lines, $\pi_2 \neq 0$ forcing monopoles — are the rest of Kibble’s
 403 classification and Mathlib’s homotopy-group library would support at least
 404 the next one; they are available and not taken, because the argument this
 405 paper runs uses domain walls and nothing else. The witness is the double
 406 well of `Examples.lean` §11.1, where the vacuum manifold is *computed* to be
 407 exactly $\{-1, 1\}$ (`wellVacuum_eq`) rather than merely known to contain both
 408 minima, and its disconnection is proved in the form the theorem consumes:
 409 no preconnected subset of it contains both, since a preconnected subset of $\mathbb{R}$
 410 is an interval and an interval spanning -1 and 1 contains 0 , which is the top
 411 of the barrier. `wellV_domain_wall` then quantifies over *every* continuous field

412 on *any* connected substrate that reaches both minima, with no formula for
 413 the field anywhere in the statement. That the class is non-empty is checked
 414 separately, because a theorem quantified over all fields is worthless if none
 415 satisfies its hypotheses: the clipped identity on $\mathbb{R}$ does, and the wall it is
 416 forced to have sits at the origin (`kink.zero.notMem`).

417 **5 Derivation 2: The Thermodynamic Con-** 418 **straint**

419 This newly formed topological boundary is subjected to gauge perturbations
 420 from the “outside” environment. Because the internal phase space is finite
 421 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
 422 and must overwrite old internal states.

423 By Landauer’s Principle (Landauer, 1961), the erasure of physical in-
 424 formation irreversibly decreases internal entropy, mandating that the sys-
 425 tem exhaust heat into the environment. Philosophers of physics (such as
 426 John Norton (Norton, 2005) and Orly Shenker (Shenker, 2000)) have cri-
 427 tiqued Landauer’s principle, arguing that physicists conflate *logical* erasure
 428 (a many-to-one mapping in abstract computational states) with *thermody-*
 429 *namic* erasure (a many-to-one mapping in physical phase space). They argue
 430 that logical erasure could conceptually be implemented by an injective phys-
 431 ical process without dissipating heat.

432 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)

433 addresses this critique by defining erasure as a non-injective mapping on the
 434 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
 435 tion of the bipartite universe (System + Bath) as strictly injective, repre-
 436 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied
 437 to abstract logical erasure, but is a mandatory mathematical consequence
 438 of any local physical state space contraction under globally reversible dy-
 439 namics. If a cognitive system’s internal topology forces a contraction of its
 440 physical state space to accommodate structural resonance, heat dissipation
 441 becomes a thermodynamic requirement. The finiteness premise enters at ex-
 442 actly this point. What a modeller can usually check about a physical update
 443 is not that it is many-to-one but that some state has become unreachable;
 444 on a finite phase space those are the same condition, and `finite_phase_`
 445 `space_dissipates` is the resulting statement. That identification preserves
 446 the Norton–Shenker distinction rather than blurring it: an update that keeps
 447 the incoming perturbation on the record is injective and dissipates nothing
 448 (`keepRecord_injective`), while one that refreshes the register holding it is
 449 not (`absorbStep_is_erasure`). Losing *which* history occurred is forced by
 450 counting alone and is free; overwriting the register that held it is what is
 451 charged $k_B T \ln 2$. A stable boundary thus constitutes a dissipative structure
 452 (Prigogine & Stengers, 1984), operating as a non-equilibrium steady state
 453 that consumes free energy to pay the thermodynamic cost of erasure. Contin-
 454 uous thermal fluctuations play a constructive role in establishing boundaries
 455 between states while contributing to the system’s flexible, adaptive response.

456 6 Derivation 3: Prediction as the Thermody- 457 namic Cost of Memory

458 A dissipative structure is driven by its environment. Write X_t for the state
459 of the structure and S_t for the environmental signal driving it, and let the
460 signal evolve by a dynamics of its own, $S_t \rightarrow S_{t+1}$, on which the structure
461 does not act. Two quantities are then defined at each step: the *memory*
462 $I(X_t; S_t)$, the mutual information between the structure's present state and
463 the signal's present value, and the *predictive information* $I(X_t; S_{t+1})$, which
464 is what that same state says about the signal's next value.

465 **What this section used to claim, and why it is withdrawn.** Earlier
466 versions of this derivation argued as follows. Model the environment as a
467 stochastic driving force $P(ext)$ and the structure's internal transitions as
468 $Q(int)$; entropy production σ is then bounded below by $D_{KL}(P||Q)/\Delta t$;
469 the principle of least action drives σ downward; therefore $D_{KL}(P||Q) \rightarrow$
470 0, and the internal transition probabilities come to mirror the statistical
471 symmetries of the environment. The final step does not follow. Rearranged,
472 the bound reads $D_{KL}(P||Q) \leq \Delta t \sigma$, so descending on σ lowers a *bound* on
473 the divergence; but a driven system does not relax to $\sigma = 0$. It relaxes to
474 a non-equilibrium steady state at strictly positive entropy production, and
475 the bound then yields only $D_{KL} \leq \Delta t \sigma_{\text{NESS}}$, a positive number. Nothing
476 forces the divergence to zero. The inequality itself is true, is formalized as an

instance obligation, and does real work — combined with Gibbs’ inequality
it gives $\sigma \geq 0$ — but the conclusion drawn from it was not licensed, and
it was that conclusion which supplied this framework’s entire connection to
predictive processing.

The replacement is a published theorem. Still, Sivak, Bell and Crooks
(Still et al., 2012) prove that for a system driven by an environmental signal
the work dissipated over one step of the drive obeys

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (1)$$

The bracket is the *nonpredictive* information: what the structure retains
about the signal’s present that says nothing about the signal’s future. Dis-
sipation is bounded below by it. Memory that fails to predict is not merely
useless to the structure; it is thermodynamic waste, and the waste is quan-
tified in energy.

Three consequences, and they are theorems rather than readings.
The bound is carried in Lean as a class field, `PredictiveDissipation`.
`still_bound (Phase3 PredictiveThermodynamics)`, for the same reason ev-
ery other physical postulate here is: its `axiom` form, quantified over all joint
laws with the dissipated work fixed, is refutable, and `Examples §18.4` refutes
it. What is *derived* is everything downstream.

First, *the bracket is non-negative*, and this is proved rather than assumed.

496 If the structure does not act on its environment, then $X_t \rightarrow S_t \rightarrow S_{t+1}$
 497 is a Markov chain and the data processing inequality gives $I(X_t; S_{t+1}) \leq$
 498 $I(X_t; S_t)$: a system cannot know more about the signal's future than about its
 499 present (`predictiveInfo_le_mutualInfo`). Mathlib carries the Kullback–
 500 Leibler divergence but no mutual information, so the latter is defined here
 501 as the divergence of the joint law from the product of its marginals, which is
 502 the standard definition and makes the library's data processing lemma apply
 503 directly.

504 Second, *the second law follows* instead of being assumed. Dissipation is
 505 bounded below by a quantity now known to be non-negative, so $W_{\text{diss}} \geq 0$
 506 (`dissipatedWork_nonneg`). This mirrors the role Gibbs' inequality plays for
 507 the older bound, with one difference that matters: Gibbs' inequality com-
 508 pares a system to an arbitrary reference distribution, whereas the data pro-
 509 cessing inequality compares the system to its own future, which is a quantity
 510 selection can act on.

511 Third, and this is the point of the replacement, *the inference now runs*
 512 *in the valid direction*. The quantity of interest sits on the small side of
 513 the inequality: $I(X_t; S_t) - I(X_t; S_{t+1}) \leq W_{\text{diss}}/k_B T$. No limit is required.
 514 Any bound on dissipation is immediately a bound on wasted memory, so a
 515 structure held to an energy budget is thereby held to a prediction standard:
 516 this is `nonpredictive_le_dissipation`. In the idealised quasi-static case
 517 $W_{\text{diss}} = 0$, the theorem `predictive_eq_of_no_dissipation` gives that every
 518 retained bit is predictive. This is a weaker statement than the one it replaces,

519 and it is the one that is true.

520 **The hypothesis is load-bearing, and we exhibit what breaks with-**
521 **out it.** The data processing step requires that the structure not write to
522 its environment. In the formalization this is not an assumption stated beside
523 the theorem but the shape of the evolution itself: the signal's step is applied
524 to the pair as the parallel kernel $\text{id} \parallel \kappa$, which acts as the identity on the
525 system coordinate and through κ on the signal coordinate, so no channel
526 from X_t to S_{t+1} exists. **Examples** §18.3 shows the restriction is not decora-
527 tion: a Markov kernel on the *pair* which copies the system's state into the
528 signal carries an independent two-bit law to a perfectly correlated one, rais-
529 ing mutual information from zero to a positive value. An agent that steers
530 its environment can indeed know more about that environment's future than
531 about its present, and Still's bound is not about such an agent.

532 **Both regimes are exhibited, not just described.** **Examples** §18 builds
533 two systems over one correlated two-bit joint law, differing only in the signal's
534 dynamics. When the signal is frozen, every remembered bit is a predictive
535 bit, the nonpredictive information is zero, and the instance is permitted to
536 dissipate nothing — with the memory itself proved strictly positive, so the
537 conclusion is about a system that does remember something. When the signal
538 is redrawn from its stationary law at each step, the predictive information is
539 exactly zero, the *entire* memory is nonpredictive, and the instance is *forced*
540 to dissipate $k_B T$ times its whole mutual information. Establishing that the

541 mutual information of the witness is neither zero nor infinite is where the
542 work of that section lies.

543 **An equality, where the old argument had only a bound.** Kawai,
544 Parrondo and Van den Broeck (Kawai et al., 2007) show that the average
545 dissipated work is exactly $k_B T$ times the relative entropy between the for-
546 ward and time-reversed phase-space densities. Carried here as a predicate on
547 an instance rather than as a further postulate, it replaces the abstract dis-
548 sipation field with a quantity an experiment can estimate — the statistical
549 distinguishability of a recording of the system from the same recording run
550 backwards — and yields the statement that the irreversibility of the dynamics
551 is at least the amount of memory that fails to earn its keep (`nonpredictive_`
552 `le_arrow`).

553 **What this does not establish.** It does not deliver the Free Energy Prin-
554 ciple. Still’s bound says that prediction is thermodynamically favoured; it
555 says nothing about hierarchical generative models, variational free energy,
556 or Bayesian inference, and we do not claim to have derived any of them.
557 The Free Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy,
558 2013; Clark, 2016; Spisák & Friston, 2026) and the General Resonance The-
559 ory of Consciousness (Hunt & Schooler, 2019) are consonant with what is
560 proved here, which is a weaker relation than derivation and the only one
561 we assert. Nor does it solve the symbol grounding problem (Harnad, 1990).
562 Maximal predictive information is compatible with the structure storing a

563 lossy, compressed or otherwise unrecognisable function of the signal; a state
564 that predicts is not thereby a state that means, and earlier versions of this
565 section overstated the case. What survives of that argument is narrower and
566 still worth saying: whatever internal states a physical structure holds, those
567 that do not help predict its environment are paid for in heat, so the pressure
568 toward states indexed to the environment is energetic rather than semantic.
569 We continue to call the phenomenon *structural resonance*, and there is no
570 teleological drive to “survive” in it: predicting the environment is simply the
571 cheaper of the two ways to hold a memory.

572 **7 Macroscopic Scaling: A Theorem and a** 573 **Commitment**

574 This is the chain’s fourth step, and it is deliberately not numbered as a deriva-
575 tion. The change is not cosmetic. Earlier drafts ran two steps of different
576 kinds together in a single paragraph: a coarse-graining argument, which is
577 a piece of mathematics, and the identification of the coarse-grained medium
578 with the brain’s endogenous electromagnetic field, which is an empirical claim
579 about cortex. Presenting the second as though it inherited the licence of the
580 first made it look like a deduction, and it is not one. Separated, the empirical
581 claim becomes the framework’s central commitment and the place where it
582 is most easily wrong — which is where a claim of this kind belongs.

583 **The formal half: coarse-graining is a theorem, and a**
584 **narrow one**

585 A network of discretely coupled units has a coupling energy that is a sym-
586 metric double sum over pairs. `Phase2_MeshConvergence.lean` proves that
587 this sum converges to a continuous integral: along a sequence of regular
588 triangulations of a region S whose edge regions shrink to points, the dis-
589 crete energy $\frac{1}{2} \sum_u \sum_v \mu(R_{uv}) f(x_u)$ converges to $\int_S f d\mu$ (`mesh_refinement_`
590 `convergence`), witnessed on the uniform partition of $[0, 1)$ in `Examples §6`.
591 The identification that makes this a coarse-graining rather than an arbitrary
592 Riemann sum is `sum_sum_mul_of_symm`: because the summand is symmetric,
593 the double sum is exactly the midpoint rule over unordered edges, so the
594 discrete energy is a quadrature of the continuous one rather than a different
595 quantity that happens to converge to it.

596 What this establishes is worth stating exactly, because it is less than the
597 phrase “coarse-graining, governed by the Renormalization Group” suggests.
598 It is a statement about an *energy functional*, not about a dynamics: noth-
599 ing here shows that trajectories of the discrete Kuramoto system converge
600 to solutions of a field equation, and the earlier appeal to the Renormaliza-
601 tion Group was decorative — no renormalization group flow is constructed
602 anywhere in this development. The limit object is a scalar density on the sub-
603 strate, in the intended reading the magnitude of the stress-energy tensor, and
604 it is not an electromagnetic field. The continuous object the later sections

605 use, `ContinuousNeuralField`, is *posited* as a structure with a coupling kernel
 606 K ; no theorem produces it from the discrete system. The gap between the
 607 finite Kuramoto model and its continuum limit is the propagation-of-chaos
 608 problem, which we have not addressed and record as open.

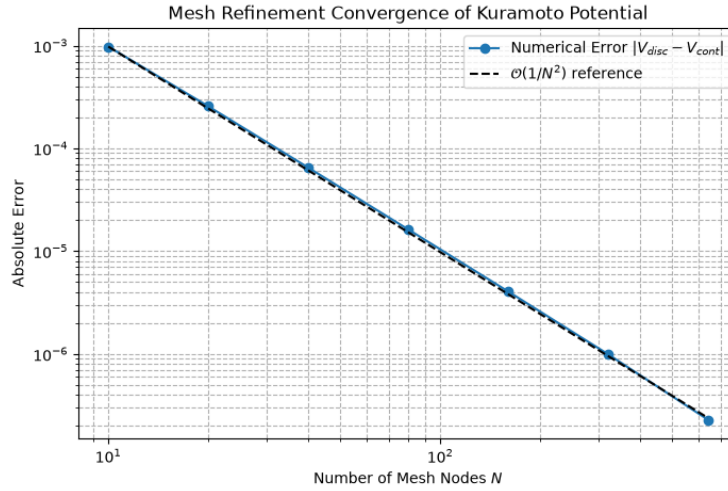


Figure 2: The coarse-graining theorem, exercised numerically. The discrete coupling energy of N units evenly spaced on the ring, with a Gaussian kernel of periodic distance and phase field $\theta(x) = \sin x$, is compared against the continuous integral it is proved to converge to. The error falls as $\mathcal{O}(N^{-2})$, the rate of the midpoint rule, which is the quadrature the Lean statement uses. The theorem asserts convergence, not a rate; the rate is what the simulation adds, and it is the ordinary rate for this quadrature rather than anything the framework predicts. The convergence is of an *energy functional* — nothing here says the discrete trajectories converge to the continuous ones.

609 **The empirical half: which physical field realizes the ker-**
610 **nel**

611 The formal half says that a large network of symmetrically coupled units
612 admits a continuum description with a symmetric kernel. It says nothing
613 about which physical field in cortex that kernel describes. Our answer is an
614 empirical commitment, and we state it as one:

615 *In the mammalian cortex, the field realizing the coarse-grained*
616 *coupling kernel is the endogenous electromagnetic field.*

617 This is a claim about biology, not a consequence of anything above it.
618 It is what makes the framework falsifiable and it is what we expect to be
619 attacked.

620 **What supports it.** Neurons do not only signal through synapses: trans-
621 membrane currents generate extracellular fields that shift the membrane po-
622 tentials of neighbouring cells, a mechanism known as ephaptic coupling. En-
623 dogenous fields of 1–5 mV/mm have been shown *in vitro* to entrain spike
624 timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015); realistic
625 head models put mesoscopic ephaptic effects within the range where they
626 can bias network dynamics (Ruffini & Santarnecchi, 2020); cable-theoretic
627 treatments show field coupling is sufficient for coincidence detection in the
628 auditory brainstem (Goldwyn & Rinzel, 2016); and stable electric fields have
629 been argued to carry information that survives representational drift in the

630 underlying spiking population (Pinotsis & Miller, 2022, 2023a,b; Cunha &
631 Lima, 2024). The field is spatially continuous and superposes contributions
632 from large populations, which is the property the framework needs: it is a
633 medium in which spatially separated activity is combined without a further
634 mechanism for combining it. This is the integrative role the CEMI field
635 theory assigns it (McFadden, 2002, 2020; Pockett, 2012).

636 **What the evidence does not support, and a claim we withdraw.**

637 Earlier drafts asserted that *the continuous macroscopic EM field is the pri-*
638 *mary dissipative structure interacting with the universe.* The word “primary”
639 is not supported by any of the work cited above and we withdraw it. What
640 the experiments show is *modulation*: Anastassiou and Koch report that en-
641 dogenous fields of 1–5 mV/mm shift spike timing by one to three millisec-
642 onds (Anastassiou & Koch, 2015), an effect that is real, reproducible, and
643 subthreshold. It biases when a cell fires; it does not decide whether it fires.
644 The corrected claim is that the endogenous field is *modulatory in its effect on*
645 *individual neurons and integrative in its spatial reach* — weak per synapse,
646 but the only variable in the system that is defined everywhere at once. The
647 framework needs the second property, not the first, and conflating them was
648 the error.

649 **Its critics.** The objection with the longest history is that the fields are
650 too weak to matter, and it has been pressed from within the field-theoretic
651 camp: Pockett, who has herself defended an electromagnetic theory of con-

652 sciousness (Pockett, 2012), catalogues its difficulties (Pockett, 2002) — the
 653 measurement problem of relating gross field structure to reported experience,
 654 the absence of a constant relationship between the two, and the question of
 655 whether the field does causal work or merely reflects it. The quantitative case
 656 for scepticism has strengthened since. Vöröslakos and colleagues measured
 657 intracranial fields during transcranial stimulation in rats and human cadav-
 658 ers and found that the great majority of applied current is shunted through
 659 the scalp, so that conventional protocols deliver fields at or below the 1 V/m
 660 range in cortex (Vöröslakos et al., 2018). That is a bound on *applied* fields
 661 rather than endogenous ones, but it constrains how much causal work a field
 662 of that magnitude can be doing, and it is the reason we do not claim the
 663 field drives spiking.

664 **What would falsify it.** We state the failure conditions rather than leaving
 665 them to a referee.

666 1. *Epiphenomenality.* If the endogenous field can be cancelled or shunted
 667 — by a conductive medium that nulls the extracellular potential with-
 668 out altering transmembrane currents — and neither behaviour nor re-
 669 port changes, the identification fails. This is the decisive experiment
 670 and it is, in principle, available.

671 2. *A magnitude gap.* If the spatial integration the framework requires
 672 demands field effects an order of magnitude above the 1–5 mV/mm
 673 measured in cortex, the commitment fails on arithmetic. The relevant

number is not whether ephaptic coupling exists but whether its coupling strength suffices to place the system above the critical coupling K_c of Derivation 7.

3. *A timescale mismatch in the wrong direction.* The prediction of Section 11 turns on the field’s dynamics being fast relative to the astrocytic and clearance variables. If recovery from sleep inertia tracks an exponential set by the electrical timescale rather than the delayed sigmoid the slow variable predicts, the framework loses the one prediction that is its own.

4. *Dissociation of field coherence from reported unity.* If large-scale field coherence is preserved across states in which experiential unity is lost — or absent in states where it is retained — the identification of the global section with unified experience has no physical carrier.

A framework that survives all four is not thereby true. It is merely still standing, which is more than most theories of consciousness arrange to be able to lose.

Geometric frustration, and the regime our theorems actually cover

Earlier drafts grounded the system’s memory capacity in *geometric frustration*: a highly coupled network cannot bring all its nodes to their lowest energy simultaneously, producing a jagged landscape of near-degenerate states

695 with the capacity to store many patterns. That argument is standard and it
 696 is not what this development formalizes. Confronting the discrepancy is the
 697 honest course, so we do it here rather than in a footnote.

698 Frustration requires competing couplings — in the Kuramoto setting, a
 699 sign structure that no phase assignment can satisfy on every edge at once.
 700 The formal setting admits this: `KuramotoSystem` carries $A : V \rightarrow V \rightarrow$
 701 $\mathbb{R}$, symmetric and otherwise unconstrained, so signed couplings are repre-
 702 sentable. But every theorem in this development that says anything about
 703 *where* the dynamics ends requires strictly positive coupling. `phase_locked_`
 704 `minimizes_potential` assumes $A_{ij} > 0$; `potential_min_iff_phase_locked`
 705 inherits it; the class field `ThermodynamicCover.A_pos` demands it of every
 706 instance, and hence of every conclusion of Derivation 5; and the convergence
 707 theorem of `Phase4.RotatingFrame` §7 requires the strictly stronger uniform
 708 bound $0 < a \leq A_{ij}$. A network with all-positive couplings is an unfrustrated
 709 ferromagnet whose potential has one minimum up to a global phase shift. It
 710 is the opposite of a jagged landscape, and it has no capacity to store multiple
 711 patterns at all. **The system in which memory is grounded and the**
 712 **system our theorems describe are not the same system.**

713 The division is not arbitrary, and it is worth recording where it falls. What
 714 survives with no assumption on the sign of the coupling is the dynamical half:
 715 well-posedness on all of $\mathbb{R}$ (`is_kuramoto_trajectory_exists`, `_unique`), co-
 716 variance under passage to a rotating frame, monotone descent of the potential
 717 (`dynamic_potential_antitone`), finiteness of the dissipation integral, and —

718 through Barbalat’s lemma — that the motion stops along every trajectory
 719 (`velocity_sq_tendsto_zero`). *That* the motion stops is frustration-agnostic.
 720 *Where* it stops is not, and every claim this framework makes about the des-
 721 tination lives on the positive side of that line. On the continuum side the
 722 accounting differs slightly: `phase_locked_minimizes_entropy_of_symm` as-
 723 sumes only that the kernel is symmetric and so tolerates signed kernels, but
 724 it is conditional on a dynamically phase-locked field existing — which is
 725 precisely what frustration may prevent.

726 We therefore scope the claim rather than repeat it. The memory capacity
 727 of a frustrated cortical network is not supported by any theorem here, and we
 728 do not assert it as part of the chain. What the chain uses is the unfrustrated
 729 regime, in which synchronization is reached and the global section exists.
 730 Whether a frustrated network can hold many near-degenerate patterns *and*
 731 still synchronize enough of itself to admit a global section is an open question
 732 about mixed-sign oscillator networks, and it is the most substantive gap we
 733 know of in this framework. Recording it as such costs a claim we would have
 734 liked to make and is preferable to holding two incompatible pictures at once.

735 **The supporting physiology**

736 The continuous medium is physically supported by astrocytic syncytia—non-
 737 neuronal networks joined by gap junctions that provide macroscopic cytoplas-
 738 mic continuity for energy and ion flows (Verkhatsky & Nedergaard, 2018).
 739 Managing the thermodynamic cost of this state requires sophisticated micro-

740 bioenergetics. Dynamic mitochondrial networks function as deep-level Mi-
 741 tochondrial Information Processing Systems (MIPS), transducing biological
 742 and environmental information to sustain the coherence of the macroscopic
 743 field (Picard & Shirihai, 2022). Discrete synaptic firings, coupled with mito-
 744 chondrial micro-mechanics, generate and sustain the field. Because the exter-
 745 nal environment is continuous, achieving maximal thermodynamic structural
 746 resonance is optimally realized within this continuous macroscopic medium
 747 (van Bree et al., 2025).

748 **8 Derivation 5: Global Sections and the Unity** 749 **of Consciousness**

750 The system is now characterized by a continuous macroscopic electrodynamic
 751 field generated by millions of coupled micro-components, forced to resonate
 752 with the environment. Following Section 7, the coupling regime assumed
 753 from here on is the unfrustrated one — all couplings positive — which is
 754 what every theorem below requires and what the frustrated network of the
 755 memory argument is not.

756 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-
 757 tors guarantees that, past a critical coupling threshold, these distributed
 758 local fields undergo a phase transition into unified synchronization (Bauer et
 759 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
 760 phase-locked state minimizes the Lyapunov potential of the system, in the

761 precise sense discussed in Section 2.1: the functional minimized is the cou-
 762 pling potential read in the frame rotating with the natural frequency, which
 763 `Phase4.RotatingFrame.lean` shows is the honest reduced description of a
 764 system of identical oscillators.

765 This phase transition mathematically glues local synchronized states into
 766 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
 767 The EM field acts as a single, indivisible topological space, providing the
 768 *Unity of Consciousness*—a necessary condition for integrated experience that
 769 solves the binding problem. While Global Workspace Theory (Baars, 1997;
 770 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
 771 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
 772 our derivation provides a continuous physical mechanism. The contrast is
 773 worth drawing carefully, because both theories are addressing the same ex-
 774 planandum from different directions.

775 We state the scope of the Lean result precisely, because it is narrower
 776 than the prose above. `ThermodynamicCover` demands that the cover’s phase
 777 field minimize the reduced Kuramoto potential, and `potential_min_iff_`
 778 `phase_locked` characterizes those minimizers exactly: they are the phase-
 779 locked configurations, in both directions. Every instance is therefore already
 780 phase-locked (`ThermodynamicCover.phase_locked`), and the class’s single
 781 condition on its local data — that patches at a common phase agree where
 782 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
 783 sections compatible. `probability_glue_unique`, which is the sheaf property

784 of the probability presheaf and contains no physics at all, then glues that
785 family into exactly one global section.

786 An earlier version of this structure proved less than the prose claimed,
787 and we record the correction because it changed what the theorem says
788 rather than how it is stated. `LocalSectionSynchronization` used to carry
789 a *global* section as a class field, together with a second field declaring every
790 local section to be a restriction of it. The global object therefore existed
791 before the sheaf condition was ever consulted, and `global_section_from_`
792 `thermodynamics` could do no more than rule out competitors to an object
793 the instance had supplied. Both fields are gone. Every field of the class
794 now mentions a single patch or a single overlap; the global section is con-
795 structed (`ThermodynamicCover.invariantMeasure`); and the removed field
796 survives as a *theorem* about that construction (`ThermodynamicCover.sync_`
797 `to_section_eq`), so the local sections are restrictions of one common measure
798 because the sheaf condition says so, not because an instance was required to
799 name one.

800 `Examples.lean` §14 is the witness that the difference is not notational.
801 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
802 the three sites, agreeing only at the site they share, where both put mass 1
803 — which is what discharges the overlap condition, by computing two num-
804 bers rather than by construction. The section the theorem returns has pro-
805 file $(2, 1, 3)$: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_`
806 `rightW`), and neither of them restricts correctly to both patches (`leftW_`

807 `glues_nothing, rightW_glues_nothing`). The global state carries informa-
 808 tion that no single patch had. `Examples.lean` §13, built the older way
 809 through the constructor `LocalSectionSynchronization.ofInvariantMeasure`,
 810 is retained alongside it for the negative result it carries: its phase field is not
 811 the constant function — the second patch a full turn ahead of the first — and
 812 by `ThermodynamicCover.phase_locked` that is as far from constant as any
 813 instance can get, since no instance can carry phases differing by anything a
 814 measure is allowed to see.

815 Two hypotheses carry physical content, and until recently each was an
 816 obligation on every instance. One still is: that synchronized patches agree
 817 on their overlaps is a modelling assumption, not a consequence of any dy-
 818 namics we formalize. The other, `thermodynamic_equilibrium`, asks that the
 819 cover already sit at the potential minimum, and no theorem says an *arbi-*
 820 *trary* system’s dynamics takes it there. “A thermodynamic phase transition
 821 into unity” is therefore still doing causal work that no general Lean result
 822 supports, though it is no longer entirely without one: `Examples.lean` §15
 823 exhibits a two-oscillator trajectory that starts a quarter turn out of phase
 824 and converges to the global minimum of the dynamic potential, with the or-
 825 der parameter tending to 1. That trajectory is not an isolated construction:
 826 `kuramoto_cauchy_problem` proves the Kuramoto initial value problem well
 827 posed on all of $\mathbb{R}$, so through every initial state there passes exactly one tra-
 828 jectory, and the exhibited one is the only solution through its own. Nor is its
 829 settling down peculiar to it: along every trajectory of a zero-frequency sys-

tem the dynamic potential converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is
 finite and equal to the potential's total drop (`dynamic_potential_tendsto`,
`dissipation_integral_tendsto`). The motion also stops: a finite integral
 does not by itself force its integrand to zero, but the dissipation rate is
 Lipschitz in time, so Barbalat's lemma applies, and every trajectory satis-
 fies $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_sq_tendsto_zero`; Mathlib has no Barbalat, so it
 is proved here as `tendsto_zero_of_lipschitz_of_integral_le`). That last
 step — locating where the motion stops — is now taken, though condition-
 ally, and by a different route than we had recorded. We had marked it as
 blocked on the LaSalle invariance principle, which Mathlib does not have,
 and on the absence of a compact invariant set in the state space $V \rightarrow \mathbb{R}$.
 That was the wrong diagnosis. LaSalle is the tool for a system with a de-
 creasing functional and nothing else; this one has more. The flow is a gra-
 dient flow of the dynamic potential, so $\dot{V} = -\sum_i \dot{\theta}_i^2$ is an identity rather
 than an inequality; the minimisers of V are characterised exactly; and on
 the region where every pair of phases lies within a quarter turn a Polyak-
 Łojasiewicz estimate holds, $2a^2W \leq (\sum_{ij} A_{ij}) \sum_i \dot{\theta}_i^2$, where W is the poten-
 tial's excess above its minimum and $a > 0$ is a lower bound for the coupling
 (`lojasiewicz_estimate`). Its proof is two extremal indices and the iden-
 tity $\sin^2 D = (1 - \cos D)(1 + \cos D)$: the site carrying the largest phase is
 pulled backwards at rate at least $a \sin D$, where D is the spread, while the
 excess is at most $\frac{1}{2}(\sum_{ij} A_{ij})(1 - \cos D)$. The compactness objection dissolves
 as well, because the invariant region is produced by the Lyapunov function

853 rather than assumed: a single pair contributes $\frac{1}{2}a(1 - \cos)$ to the excess, so
 854 an excess below $a/2$ forces every pairwise cosine positive for all time, and
 855 continuity on the connected half-line then prevents any pair from chang-
 856 ing branch (`spread_le_of_small`). Descent plus the estimate integrates to
 857 $W(t) \leq W(0)e^{-ct}$ with $c = 2a^2 / \sum_{ij} A_{ij}$ (`excess_decay`), the rate makes each
 858 $|\dot{\theta}_i|$ exponentially small and hence integrable on $[0, \infty)$, and a trajectory with
 859 integrable velocity converges. The conclusion is `kuramoto_tendsto_global_`
 860 `minimum`: if the phases start within a quarter turn of each other and the initial
 861 excess is below $a/2$, the trajectory converges to a phase-locked configuration
 862 which is a *global* minimiser of the dynamic potential, with order parameter
 863 $r^2 = 1$. That is the configuration `thermodynamic_equilibrium` postulates,
 864 reached rather than assumed. `Examples.lean` §17 fires it on three sites with
 865 initial data $(0, 0, \frac{1}{2})$, where the Kuramoto system has no closed-form solu-
 866 tion: the trajectory comes from the existence theorem and its limit from the
 867 convergence theorem, so neither is written down.

868 `Phase5_EquilibriumBridge.lean` closes the gap that remained between
 869 the two results. `ThermodynamicCover.ofConvergentTrajectory` takes a
 870 cover whose phase field is the limit of such a trajectory and returns an in-
 871 stance whose `thermodynamic_equilibrium` field is discharged *by that con-*
 872 *vergence*; the cover's coupling matrix is the coupling matrix of the dynamics
 873 rather than a separately chosen one, so the system whose equilibrium the
 874 class asserts and the system whose trajectory is run are the same system.
 875 `Examples.lean` §17.1 is the resulting witness: three patches over the three-

876 site substrate, each carrying its own mass profile on its own open set, above a
 877 phase field that no line of the construction writes down. The trajectory that
 878 reaches it starts at $(0, 0, \frac{1}{2})$, which is provably not phase-locked (`trioCover_`
 879 `start_not_locked`), so on this cover the configuration Derivation 5 requires
 880 is arrived at rather than posited. The state the three patches glue to has
 881 profile $(2, 1, 3)$ and is none of the three profiles the instance carries (`trioW_`
 882 `ne_glued`), so one cover now carries both the emergence reading of the gluing
 883 and the derived reading of the equilibrium. What is still assumed on it is the
 884 overlap agreement, and nothing else.

885 The two hypotheses cannot be dropped, and this is the honest scope of
 886 the result. Splay and twisted configurations are equilibria of the same flow,
 887 so “every trajectory reaches the unified state” is false as a general statement,
 888 and the initial-data condition is the standard arc condition under which the
 889 true statement holds. What is claimed is therefore conditional — unity is
 890 reached from configurations already sufficiently coherent — and not that an
 891 arbitrary initial state is carried into unity. What Lean establishes about the
 892 gluing itself is the other half: *given* locally synchronized patches, a global
 893 state exists, is unique, and is produced by the gluing rather than presupposed
 894 by it.

895 Integrated Information Theory begins from phenomenological axioms and
 896 derives a scalar Φ measuring irreducibility of a system’s cause-effect structure,
 897 with the exclusion postulate selecting a single maximum- Φ complex as the
 898 locus of experience. Our sheaf-theoretic global section plays a structurally

899 analogous role — it too picks out a distinguished unified object — but it
900 is arrived at from the opposite end: not postulated from the character of
901 experience, but forced by a thermodynamic phase transition in a physical
902 field. This buys two things IIT does not have. First, the unity condition
903 is checkable from field dynamics rather than requiring the computation of
904 Φ , which is intractable for systems of realistic size and whose definition has
905 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
906 out rather than being postulated: gluing either succeeds on a given cover or
907 it does not, so there is no separate principle needed to rule out overlapping
908 conscious subsystems. It also inherits IIT’s exposures. Critics have argued
909 that Φ can be high in systems with no plausible claim to experience, such
910 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable
911 to the mirror objection, since a physical system could in principle satisfy
912 the gluing condition without being cognitively interesting. We take this
913 seriously rather than dismissing it: our reply is that gluing is necessary but
914 not sufficient, and that the topological boundary of Derivation 1 and the
915 reflexive structure of Derivation 6 are additional required conditions. IIT, by
916 contrast, treats Φ as both necessary and sufficient.

917 Global Workspace Theory faces a complementary difficulty. It specifies
918 a functional architecture — a limited-capacity workspace broadcasting to
919 specialized modules — without specifying what physical property of broad-
920 casting makes the broadcast content conscious, leaving the theory open to the
921 charge that it explains access consciousness while leaving phenomenal con-

consciousness untouched (Block, 1995). Our framework can be read as supplying the missing physical substrate for the workspace: the continuous macroscopic field *is* the medium in which integration is realized, and the $K_c = 2D$ threshold gives a physical condition for ignition rather than a functional stipulation of it. Where GWT is agnostic about implementation, we make a falsifiable commitment — unity requires a continuous physical medium, which predicts that a functionally equivalent discrete broadcast architecture would not be conscious. We regard this as a genuine cost of the position, not a free advantage: it is the point at which our framework departs most sharply from computational functionalism and where it is most readily falsified. Contrast this derivation with panpsychism: consciousness is not assumed as a fundamental property of matter. It requires the initial existence of a topological defect to establish a bounded “inside” (Derivation 1), followed by a thermodynamic phase transition into unity that occurs when integrated spatial coupling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both the foundational topological boundary and the subsequent mathematically defined phase transition, the system remains a fragmented, non-conscious collection of micro-states. A unified field of perception, however, is not yet a “Self.”

941 9 Derivation 6: Reflexive Topology and the 942 Emergence of the Self

943 While a global section provides Unity, a “Self” requires reflexivity. The
944 unified EM field is a dissipative structure that must resonate with external
945 perturbations while managing its own internal phase transitions. Time evo-
946 lution naturally drives the field to minimize thermodynamic friction with its
947 own internal state-changes.

948 This mandates *auto-resonance*: the field must predict itself. The “Self”
949 emerges as a higher-order topological feature when the internal geometry
950 folds to include a low-dimensional avatar of its own computational and causal
951 boundary within its predictive model (`Phase6_ReflexiveTopology.lean`)
952 (Levin, 2019). The system is unified, predictive, and explicitly maps its
953 own distinct existence from the environment. This architecture provides the
954 physical foundation for a self-aware consciousness, bounded by the continuous
955 measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).

956 **What carries the weight.** `reflexive_topology_implies_self` is the
957 Banach fixed-point theorem with the physical commitments in its hypothe-
958 ses: the space of global sections is assumed nonempty, metric and complete,
959 and the predictive map is assumed to contract. `Examples.lean` §10 dis-
960 charges all four on the three-site substrate of Derivation 5, and the metric
961 is no longer a placeholder. The section first pierces the sheafification: on

962 a discrete substrate every point has a smallest open neighbourhood, so the
 963 mass a germ carries at a point is well defined, and `massEquivOn` proves that
 964 over *any* open the sections of the probability sheaf *are* the finite measures
 965 on that open, read as densities against counting measure — the global case
 966 being $U = \top$. Mathlib supplies no inverse to this sheafification — its `TopCat`
 967 sheafification has neither an adjunction nor an `isIso_toSheafify` — so the
 968 correspondence had to be constructed. The metric on sections is then the
 969 uniform distance between the densities of the measures they glue to — equiv-
 970 alent to their total variation on a finite substrate, though not equal to it —
 971 and it is proved complete by exhibiting the limit section of a Cauchy sequence
 972 of densities, not by the degenerate route of showing Cauchy sequences are
 973 eventually constant. An earlier version of this witness used the 0/1 metric,
 974 under which every contraction is constant; that theorem is kept in the source
 975 as the record of why the earlier witness was empty.

976 **The map is the avatar, or there is no map.** Until recently the weight
 977 was not carried where the section claimed. `ReflexiveBoundary` carried the
 978 predictive map as free data sitting beside the avatar, and the fixed-point
 979 theorem projected that map out and never touched the avatar; so replacing
 980 the avatar by a constant left a legal boundary with the same predictive model
 981 and the same Self. We recorded that as a theorem rather than a caveat, and
 982 it was the section’s central defect: a fixed-point theorem that cannot detect
 983 whether the avatar reads the field is Banach’s theorem with a consciousness-

984 flavoured label. The structure is now built the other way round. A reflexive
 985 boundary carries the avatar's *write* — `auto_resonance`, the encoding of the
 986 global state into the avatar region — and its *read-out*, the global state that
 987 encoding predicts; the predictive map is their composite and is not a field.
 988 The self-model therefore runs through the avatar region or it does not exist.
 989 Three consequences follow that could not previously be stated. The Self lies
 990 in the range of the read-out (`self_mem_range_readout`), so it is reconstructed
 991 from a single section over the avatar region: the low-dimensional fold this
 992 section has always asserted, as a theorem. Under the resonance condition
 993 below, the Self satisfies $s = \text{readout}(s|_{\text{avatar}})$ (`self_eq_readout_restrict`) —
 994 the field is what its own localized self-encoding reconstructs. And blinding
 995 the avatar now *moves* the Self: the blinded boundary's map is constant,
 996 its unique fixed point is the state its stipulated contents extend to, and
 997 the original Self is not a fixed point of it (`not_self_of_constResonance`,
 998 witnessed on the cortex by `cortexBlind_self_ne`). The theorem that used
 999 to say the opposite no longer typechecks, because a contraction hypothesis
 1000 about a boundary's own map says nothing about the blinded one's.

1001 **Auto-resonance as a constraint.** That the avatar's state *is* the presheaf
 1002 restriction of the global state is stated as a predicate, `IsRestriction-`
 1003 `Resonance`, rather than added to the structure, so that results say which
 1004 of them need it. In the positive direction, if a family of resonant avatars *cov-*
 1005 *ers* the substrate then the global state is determined by their local readings

1006 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but
1007 recovered from them, by the same sheaf condition that Derivation 5 uses.
1008 In the negative direction a constant avatar fails the predicate as soon as the
1009 avatar region separates two states of the field. Both halves are witnessed on
1010 the three-site substrate, where that separation is proved before anything is
1011 claimed for it, so the condition is a constraint there and not a formality.

1012 **The contraction rate is K and D , not a numeral.** The remaining stip-
1013 ulation was the rate. “The predictive map contracts by $1/2$ ” was a fact about
1014 a map we had defined to average with a baseline, not a claim about a sub-
1015 strate. The rate is now $\exp(-(K - K_c)\tau/2)$ with $K_c = 2D$ (`resonanceRate`)
1016 — the linear relaxation of the mean-field order parameter over an interval τ ,
1017 read off the same threshold Derivation 7 (Section 10) proves the phase tran-
1018 sition at. Two theorems then split the parameter plane. Above threshold
1019 the rate is below one and Banach applies, so the Self exists and is unique
1020 (`self_of_supercritical`). At or below threshold the rate is at least one,
1021 and `ContractingWith` at that rate is false *for every self-map of every metric*
1022 *space* (`not_contractingWith_resonanceRate`): the argument is not merely
1023 unproved below K_c but unavailable. We are careful about what this does and
1024 does not say. It does not say a Self fails to exist below threshold — only that
1025 this route to one is closed — and the Lipschitz bound at that rate remains
1026 an instance obligation, discharged on the three-site witness at $K = 3$, $D = 1$,
1027 $\tau = 2 \ln 2$, where the rate is exactly the one half the read-out achieves. What

1028 changed is that the obligation is stated in the substrate's own parameters
 1029 and its consequence is conditional on the threshold, rather than on a con-
 1030 stant chosen to make the witness work. The link to Derivation 7 is carried
 1031 by a theorem rather than by a shared numeral, and the distinction is worth
 1032 being exact about. `self_of_supercritical` takes $K > K_c$ as its hypothesis;
 1033 K_c is a definition unfolding to $2D$, so that statement is insensitive to whether
 1034 the bifurcation is proved at all. `Chain.lean` states the link in the form that is
 1035 not. `self_of_coherent_order_parameter` takes the *existence of a coherent*
 1036 *order parameter* as its hypothesis and recovers the threshold inequality from
 1037 it, by `supercritical_of_coherent` — the contrapositive of the subcritical
 1038 uniqueness result of Section 10, since a strictly positive solution of the self-
 1039 consistency equation cannot exist at or below K_c . Derivation 6's fixed point
 1040 therefore rests on Derivation 7's theorem.

1041 **What is still not established.** That any field dynamics produces this
 1042 particular read-out or performs the restriction; and that this three-site sub-
 1043 strate's coupling is $K = 3$ — the parameters are a choice of units for the wit-
 1044 ness, not a measurement, and nothing here derives a coupling constant from
 1045 cortex. We also note a negative result found in the course of this work: the
 1046 Lévy–Prokhorov metric, which the natural weak-convergence reading would
 1047 suggest, does *not* support this argument on a discrete substrate, because the
 1048 ε -thickening of a set is the set itself for $\varepsilon < 1$ and the relaxation map is
 1049 consequently not a Lévy–Prokhorov contraction.

1050 10 Derivation 7: Continuous Stochastic Ther- 1051 modynamics and Field Theory

1052 To move beyond finite graph analogies, we model the macroscopic electro-
1053 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
1054 continuous spatial phenomenon, not a discrete network. We extend the
1055 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$
1056 (`Phase8ContinuousField.lean`).

1057 Subject to thermodynamic noise D , the system undergoes non-equilibrium
1058 stochastic thermodynamics. Rather than a static Landauer bound, the field
1059 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (2)$$

1060 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
1061 dergo gradient descent on this entropy production functional.

1062 That gradient descent is a theorem rather than a modelling assertion,
1063 for finite substrates. Reading σ as a functional of the coupling kernel at
1064 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative
1065 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (3)$$

1066 and `sigmaOfKernel.eq.entropy_production_rate` identifies the functional
 1067 being differentiated with `entropy_production_rate` itself whenever the sub-
 1068 strate carries counting measure. `structural_resonance_decreases_entropy_production`
 1069 then concludes that σ is non-increasing along any coupling trajectory relax-
 1070 ing down its own gradient at a positive rate c . Earlier drafts proved this
 1071 only for an abstract functional S on an abstract inner-product space with
 1072 the gradient supplied as data; the identification of that S with σ was infor-
 1073 mal, and Table S1 flagged the row accordingly. Two limitations stay explicit,
 1074 and the first is narrower than we previously reported. The derivation is for
 1075 *finite* substrates. We had recorded the obstacle to the continuum case as
 1076 differentiation under the integral sign with respect to the kernel; that was
 1077 a misdiagnosis. σ is not a general functional that happens to be written as
 1078 an integral — it is a quadratic functional of an *affine* image of the kernel,
 1079 since the drift is a fixed ω plus a linear operator applied to K , and σ is a
 1080 constant times the squared norm of the drift. A squared norm is Fréchet
 1081 differentiable on any real inner-product space and an affine map is its own
 1082 derivative, so the composite differentiates by the chain rule with no limiting
 1083 argument. `hasFDerivAt.quadratic_of_affine` states that in full generality,
 1084 and `hasFDerivAt.sigmaOfKernel_of_operator` re-derives the finite gradient
 1085 from it, showing that the closed form displayed above is a convenience rather
 1086 than the content. What the continuum case still needed was therefore one
 1087 object: the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as a bounded op-
 1088 erator $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz estimate — the integral

1089 kernel is bounded by 1, so on a finite measure space the operator norm is at
 1090 most $\mu(M)^{1/2}$. That object is now built. `kernelCLM` is the operator, for any
 1091 kernel factor bounded by 1, with `opNorm_kernelCLM_le` the stated bound;
 1092 `continuumDriftCLM` is the instance with the Kuramoto factor $\sin(\theta_y - \theta_x)$.
 1093 One step beyond the estimate is needed and is worth naming, because the cor-
 1094 rected scope above did not see it: the descent theorems ask for the derivative
 1095 as a *gradient vector*, not as an arbitrary bounded functional, and producing
 1096 one requires the adjoint, which exists because L^2 is complete. `hasFDerivAt_`
 1097 `quadratic_grad` supplies it for an arbitrary bounded operator between real
 1098 Hilbert spaces, giving $\nabla\sigma(K) = (2/D) A^*(AK + \omega)$, so the finite and contin-
 1099 uum cases are now the same theorem. `structural_resonance_decreases_`
 1100 `sigmaContinuum` is the descent result with no finiteness hypothesis in it, and
 1101 `sigmaContinuum_eq_integral` identifies the functional being differentiated
 1102 with the integral `entropy_production_rate` computes. `Examples.lean` §20
 1103 runs this on Lebesgue measure restricted to $(0, 1)$, a substrate that is neither
 1104 finite nor atomic, so the finite-substrate results of the previous paragraph do
 1105 not typecheck there at all. The second limitation is unchanged: the phase
 1106 field is held fixed throughout, so this is plasticity of the coupling at frozen
 1107 phases, not joint (θ, K) dynamics. `Examples.lean` §5 exercises the whole
 1108 chain on a two-site substrate whose coupling relaxes as e^{-t} and whose entropy
 1109 production decays as e^{-2t} , confirming that the gradient is not identically zero
 1110 and that the descent is strict rather than vacuously monotone.

1111 A second result about σ concerns where the descent is headed, and it has

1112 been strengthened since the last draft in a way that also corrects an estimate
 1113 we had recorded. That the phase-locked field minimizes σ was proved only
 1114 against competitors sharing its spatial mean drift — a restriction that left
 1115 the theorem close to Jensen’s inequality and no more — and we had judged
 1116 that lifting it would require modelling how the mean drift varies with the
 1117 competitor. That judgement was wrong. For a *reciprocal* kernel, one with
 1118 $K(x, y) = K(y, x)$, the restriction is not a restriction: the coupling integrand
 1119 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging its two sites, because
 1120 K is symmetric while $\sin$ is odd, so Fubini identifies its double integral with
 1121 its own negative and the total coupling contribution vanishes. The total drift
 1122 $\int (\omega + \text{coupling})$ is therefore $\int \omega$ whatever the phases are (`total_drift_eq_of_`
 1123 `symm`): coupling moves drift between sites but neither creates nor destroys it.
 1124 Evaluating that conserved quantity at the phase-locked configuration, where
 1125 the drift is constant, pins its value, and the mean-matching hypothesis follows
 1126 for every competitor. `phase_locked_minimizes_entropy_of_symm` is the un-
 1127 restricted statement. Reciprocity is the physically intended condition — an
 1128 ephaptic field acts on both sites of a pair alike, and it is the same symmetry
 1129 `ThermodynamicCover.A_symm` already imposes in Derivation 5 — but it is
 1130 not a formality, and we prove that rather than assert it. `Examples.lean` §16
 1131 exhibits a *one-way* coupling on two sites, in which a feels b and b does not
 1132 feel a , whose phase-locked field is strictly beaten: displacing the second phase
 1133 to $-\pi/2$ cancels the first site’s natural drift while adding none at the sec-
 1134 ond, halving σ . So “phase-locking minimizes entropy production” is false for

1135 general kernels and true for reciprocal ones, and the hypothesis cannot be
 1136 removed by working harder.

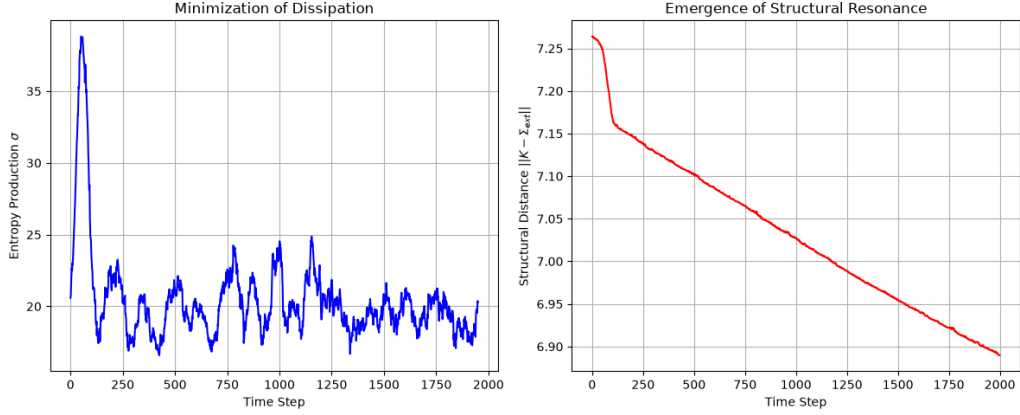


Figure 3: Coupling plasticity at frozen statistics: $N = 50$ units driven by perturbations drawn from a fixed environmental covariance Σ_{ext} , with the coupling matrix K descending the gradient of σ and clipped to stay non-negative and hollow. **Left:** entropy production, smoothed over 50 steps, falls from its transient and settles into a fluctuating plateau — a non-equilibrium steady state at strictly positive σ , not a relaxation to zero. That plateau is the reason Section 10 withdraws the old inference from $\sigma \rightarrow \min$ to $D_{KL} \rightarrow 0$: the system visibly does not reach $\sigma = 0$. **Right:** the distance $\|K - \Sigma_{\text{ext}}\|$ decreases monotonically over the run, the internal geometry drifting toward the statistics of the drive. Two cautions. The descent here is on σ at frozen phases, which is the regime the theorem covers; and the right panel falls by about 5% over 2000 steps, so what it shows is a direction of drift, not convergence.

1137 We define the critical coupling threshold $K_c = 2D$. The phase transition
 1138 to the unified “Global Section” occurs when the mean-field spatial coupling
 1139 exceeds this thermodynamic noise threshold, linking the emergence of Unity
 1140 to physical temperature and noise.

1141 Earlier drafts left $K_c = 2D$ as a bare stipulation validated only by

1142 simulation. `Phase8_SelfConsistency.lean` now proves the half of it that
 1143 is ordinary real analysis. For the noisy mean-field Kuramoto model with
 1144 identical frequencies, the stationary density at order parameter r is the
 1145 von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency requires $r =$
 1146 $R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel functions,
 1147 so we formalize that ratio directly as a ratio of integrals over $[-\pi, \pi]$. In-
 1148 tegration by parts with $u = \sin$, $v = -e^{a\cos\theta}$ gives an exact identity with
 1149 no boundary term, $\int \cos\theta e^{a\cos\theta} d\theta = a \int \sin^2\theta e^{a\cos\theta} d\theta$, so R is a times
 1150 the mean of $\sin^2$ under the von Mises weight. Bounding that mean by 1
 1151 would put the threshold at D ; the true threshold is $2D$, and the entire fac-
 1152 tor of two is the sharper bound $\mathbb{E}_a[\sin^2\theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
 1153 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
 1154 ($\theta \mapsto \pi - \theta$, then $\theta \mapsto \pi/2 - \theta$), after which the integrand is a product
 1155 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
 1156 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
 1157 `subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
 1158 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
 1159 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
 1160 ther facts about the threshold have since been proved, and they narrow the
 1161 gap between this row and a full “Theorem” to a single named statement.
 1162 First, the fold argument gives a *strict* inequality for $a > 0$ (`vonMisesC2_`
 1163 `pos`, hence $R(a) < a/2$), which settles the threshold itself: at $K = K_c$ the
 1164 self-consistency map is tangent to the diagonal at the origin but falls strictly

1165 below it thereafter, so $r = 0$ remains the only solution and the bifurcation
 1166 happens strictly *after* K_c , not at it (`fixed_point_eq_zero_of_le_critical`).
 1167 Second, factoring $R(a) = a E(a)$ shows that a non-zero r solves the equation
 1168 exactly when $E(Kr/D) = D/K$ (`coherent_iff_sRatio_eq`), so the coherent
 1169 branch is a level set of the single function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third, and using
 1170 only $E < 1/2$ together with compactness, every coherent solution tends to
 1171 zero as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`): the branch
 1172 emerges continuously from the incoherent state rather than appearing at a
 1173 finite distance from it, which is what makes the transition supercritical in the
 1174 technical sense. That statement quantifies over *every* solution, so it does not
 1175 presuppose the uniqueness it does not have. The level-set form also says ex-
 1176 actly what uniqueness is: the injectivity of E on $(0, \infty)$. That is now proved,
 1177 in the stronger form that E is strictly decreasing on $[0, \infty)$ (`vonMisesSRatio_`
 1178 `strictAntiOn`), so the level set is a single point and the coherent solution
 1179 above threshold is unique. The proof differentiates nothing. Differentiating
 1180 the exponential family in its natural parameter is the obvious route and it
 1181 fails: it costs differentiation under the integral sign and leaves a covariance
 1182 that is not sign-definite pointwise. What works instead keeps the target as
 1183 a two-point inequality, folds both integrals onto $[0, \pi/2]$ by the same two
 1184 reflections that gave the factor of two above, and then evaluates at a sin-
 1185 gle crossing point supplied by the intermediate value theorem — no product
 1186 measure and no Fubini anywhere. We give that argument in the supplement,
 1187 where it belongs; what the main line of the paper needs from it is the one

1188 sentence that E is strictly decreasing. Run with $\cos \theta$ in place of $\sin^2 \theta$ the
1189 same argument needs no fold at all — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the
1190 variable the family is tilted by — and gives `besselRatio_strictMono`: R is
1191 strictly increasing on all of $\mathbb{R}$, the monotonicity of the Bessel ratio our earlier
1192 drafts recorded as unproved.

1193 The converse half is `supercritical_fixed_point_exists`: for $K > K_c$
1194 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
1195 this to require a second-order expansion of the Bessel ratio, and it does
1196 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
1197 tion the integration by parts already supplies — the only facts needed are
1198 that E is continuous and that $E(0) = 1/2$, the latter because the weight
1199 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
1200 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
1201 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
1202 ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.
1203 The intermediate value theorem closes the gap. Continuity of E is Math-
1204 lib's continuity of a parametric interval integral with jointly continuous in-
1205 tegrand, so no dominated-convergence argument is written by hand, and
1206 the existence proof never uses monotonicity of R : the fixed point is pro-
1207 duced by the intermediate value theorem, not by iterating the map. That
1208 R is in fact strictly increasing is proved below, and no threshold result de-
1209 pends on it. `critical_coupling_is_threshold` packages the two halves,
1210 and `exhibits_phase_transition_coherent` applies the supercritical half to

1211 any substrate satisfying the predicate of the previous paragraph — so ex-
 1212 ceeding K_c now implies the existence of a coherent stationary order pa-
 1213 rameter rather than merely comparing against a named number. With
 1214 uniqueness available both are sharpened — `supercritical.fixed.point_`
 1215 `existsUnique`, `supercritical.solution.set` (the non-negative solutions
 1216 are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits.phase_`
 1217 `transition.unique.coherent` — and the two monotonicities together fix
 1218 the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing
 1219 puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
 1220 ing carries that to $r_1 < r_2$ (`coherent.branch.strictMono`). The coherent
 1221 order parameter grows strictly with the coupling, which with the continuity
 1222 at threshold is the qualitative bifurcation diagram.

1223 One more step makes the equation mean in Lean what it means on paper.
 1224 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
 1225 r with a suggestive name, and nothing in the source says it has anything
 1226 to do with an order parameter. `Phase8.SelfConsistency §8` supplies the
 1227 identification. `vonMisesDensity` is the normalized weight, proved to be a
 1228 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
 1229 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system
 1230 `order.parameter.complex` of `Phase4`; and that average is proved to be ex-
 1231 actly $R(a)$ — real, because the density is even, so the mean direction van-
 1232 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
 1233 name promises: r solves $r = R(K, r)$ if and only if the von Mises density

1234 it induces has order parameter r . Read through this, the existence theorem
 1235 says that above K_c there is a stationary density whose own order parameter
 1236 is positive, and the solution $r = 0$ is identified with the uniform density,
 1237 whose order parameter is zero.

1238 What remains not established, and we separate it carefully. The sta-
 1239 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
 1240 $(\omega + K \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
 1241 and uniqueness of its stationary solution, and a spectral stability argument,
 1242 none of which Mathlib has. Of the coherent branch we prove existence,
 1243 uniqueness, continuous emergence at the threshold and strict growth in K ,
 1244 but not that it is the dynamically selected one: nothing in that file men-
 1245 tions a trajectory, so stability and selection are not statements it can make.
 1246 And while `selfConsistency` is now connected to the order parameter of
 1247 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
 1248 integral, `order_parameter_complex` is an average over finitely many oscilla-
 1249 tors, and relating them is the mean-field limit — propagation of chaos for
 1250 the finite Kuramoto system — which is a research programme rather than
 1251 a lemma. That single gap is now the whole of what separates our threshold
 1252 predicate from a statement about dynamics. Table S1 records the row ac-
 1253 cordingly as a theorem about the self-consistency equation from an assumed
 1254 density, not as a derivation of the threshold from the dynamics. Perceptual
 1255 anomalies such as visual snow can be modeled as boundary conditions where
 1256 microscopic thermal noise (D) fails to be fully coarse-grained away by the

1257 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

1258 Clinical evidence for this phase transition is observed during general anes-
1259 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
1260 that anesthetics like propofol induce unconsciousness not by suppressing neu-
1261 ral firing, but by disrupting the phase alignment of high-frequency cortical
1262 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
1263 tural resonance of the continuous field, forcing the integrated spatial coupling
1264 below the critical threshold K_c and fracturing the unified Global Section back
1265 into disjoint, unconscious local fields.

1266 11 The Sleep-Inertia Timescale Mismatch

1267 Most of what this framework asserts is shared with some rival: that corti-
1268 cal synchrony matters, that dissipation constrains cognition, that unity is a
1269 physical rather than a computational property. One prediction is not shared,
1270 and it follows from the framework’s specific commitment — that the medium
1271 carrying the unified state is a continuous field whose effective coupling de-
1272 pends on the physical geometry of the extracellular space, not on the rhythms
1273 alone. Where that commitment differs from a purely electrical account, it
1274 differs in an observable way, and we set out the difference precisely enough
1275 to be wrong.

1276 The daily loss of consciousness during sleep is modeled here as an active
1277 physical disruption of the medium rather than a quiescence of the oscillators.

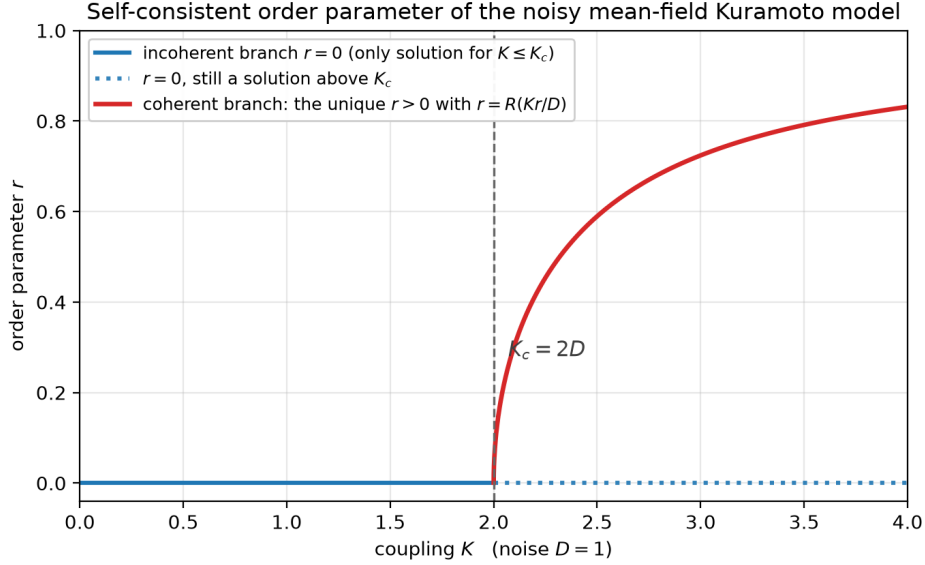


Figure 4: The self-consistency equation $r = R(Kr/D)$ of Section 10, drawn at $D = 1$. Every feature of this picture is a proved statement about the equation, and the figure exists because none of them had been shown to a reader. At or below $K_c = 2D$ the incoherent state is the *only* non-negative solution, threshold included (`fixed_point_eq_zero_of_le_critical`); above it exactly one $r > 0$ joins it (`supercritical_fixed_point_existsUnique`), so the solution set is $\{0\}$ below and $\{0, r\}$ above; that branch tends to 0 as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`), which is what makes the transition supercritical; and it increases strictly in K (`coherent_branch_strictMono`). The curve is computed by quadrature of the same integrals the Lean file defines, solving $E(a) = D/K$ for the concentration $a = Kr/D$ by bisection — legitimate precisely because E is proved strictly decreasing. It is an illustration of theorems, not evidence for them. What it does *not* show is stability: nothing in the development says the coherent branch is the dynamically selected one.

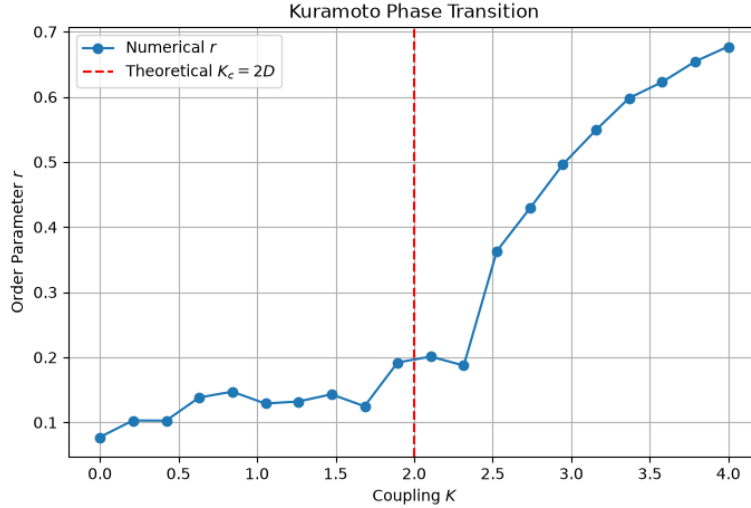


Figure 5: The same threshold reached by simulation rather than by analysis, and by a different route, which is why the two figures are shown side by side rather than merged. Here $N = 100$ oscillators with *no* noise carry natural frequencies drawn from a Lorentzian of scale $D = 1$, and the asymptotic order parameter is measured from the trajectories; the threshold for that ensemble is $K_c = 2D$ by Kuramoto’s original self-consistency argument. Figure 4 is the *noise*-driven model with identical frequencies, whose threshold is $2D$ by Sakaguchi’s (Sakaguchi, 1988), and it is the one the Lean file formalizes. The agreement of the two numbers is a known feature of the mean-field model, not a result of ours. The finite- N curve rises before the threshold and rounds off at it, as a finite sample must: $r \sim N^{-1/2}$ from incoherent phases alone.

1278 During sleep, astrocytic shrinkage expands the extracellular space, allowing
1279 an influx of cerebrospinal fluid (CSF) to flood the brain for metabolic waste
1280 clearance (Xie & Nedergaard, 2013). This influx alters the spatial medium
1281 itself, disrupting the continuous structural resonance and preventing the co-
1282 herent sheaf-theoretic gluing that conscious unity requires on this account.
1283 The oscillators need not stop for the field to fragment; the geometry they are

1284 coupled through is what changes.

1285 The prediction concerns sudden awakenings. When jolted awake, the lo-
1286 cus coeruleus floods the cortex with noradrenaline. High-frequency electrical
1287 rhythms attempt to phase-align in milliseconds (Bastos & Miller, 2021), but
1288 the noradrenaline-driven astrocytic constriction required to squeeze the ex-
1289 cess CSF out of the interstitial pathways operates on a timescale of seconds
1290 to minutes (Xie & Nedergaard, 2013) — three to six orders of magnitude
1291 slower. During that lag the electrical oscillators are synchronizing across a
1292 waterlogged topology, and the fragmented structural resonance is what is ex-
1293 perienceed as grogginess. The unified Global Section re-establishes only when
1294 astrocytic volume finishes matching the state of the electrical rhythms.

1295 **Why this implies a delayed sigmoid.** The two timescales separate
1296 cleanly, and that separation is what makes the prediction sharp rather than
1297 merely qualitative. Write $K(t)$ for the effective coupling of the continuous
1298 field, which on this account is a monotone function of the reconstricting
1299 extracellular geometry, and let $r(t)$ be the order parameter of the phase
1300 field. The electrical variable is fast: at every instant the oscillators are al-
1301 ready at the configuration appropriate to the current K , so r tracks the
1302 equilibrium branch $r(K)$ rather than relaxing on its own. The slow vari-
1303 able $K(t)$ then carries the whole time course, and it relaxes over 10^2 – 10^3
1304 seconds. What Derivation 7 establishes about that branch is exactly what
1305 the functional form needs, and each of the three pieces is a theorem rather

1306 than a picture: at or below the threshold the incoherent solution $r = 0$ is the
 1307 *only* non-negative solution of the self-consistency equation (`fixed_point_eq_`
 1308 `zero_of_le_critical`); above it the coherent solution exists and is unique
 1309 (`supercritical_fixed_point_existsUnique`); it emerges *continuously* at
 1310 K_c rather than by a jump (`coherent_branch_continuous_at_threshold`);
 1311 and it is strictly increasing in K (`coherent_branch_strictMono`). A jump
 1312 discontinuity at threshold would predict a step rather than a sigmoid, and
 1313 a non-monotone branch would predict overshoot; neither is what the mathe-
 1314 matics gives. Composing a monotone relaxation $K(t)$ with a branch that is
 1315 identically zero below K_c , rises continuously from zero at K_c , and saturates
 1316 above it yields a curve with three phases: a genuine *latency* while $K(t) < K_c$,
 1317 during which recovery is not merely slow but absent; a continuous rise as the
 1318 threshold is crossed; and a saturation as $K(t)$ reaches its waking value. That
 1319 is a delayed sigmoid, and the delay is not a fitting parameter — it is the time
 1320 the slow variable takes to reach K_c .

1321 **What the alternative predicts.** An account on which grogginess is the
 1322 electrical realignment itself — the rhythms are simply slow to re-phase —
 1323 has no slow variable and no threshold to cross. Its recovery is a relaxation
 1324 toward a fixed point, and relaxation toward a fixed point is exponential near
 1325 it: monotone from $t = 0$, fastest at the start, with no flat foot. The two
 1326 curves are distinguishable in the first seconds, which is where they differ
 1327 most, and the discriminating feature is the presence or absence of an initial

1328 interval of no recovery. This is a comparison of functional forms on data
1329 of a kind already collected in sleep-inertia studies, not a proposal for a new
1330 instrument.

1331 **What would falsify it.** An exponential fit that beats a delayed-sigmoid fit
1332 on the early portion of the recovery curve. A recovery latency that does not
1333 scale with an independent measure of extracellular volume or CSF clearance.
1334 A pharmacological manipulation of astrocytic volume that leaves the latency
1335 unchanged while altering the rhythms, or the converse. Any of these would
1336 tell against the framework’s central empirical commitment, which is not that
1337 synchrony matters but that the *medium* is what gates it.

1338 **What this prediction assumes.** Three things, and they are not small.
1339 The identification of the effective coupling K with a monotone function of
1340 extracellular geometry is a modelling assumption, not a measurement. The
1341 threshold value $K_c = 2D$ is derived from an assumed von Mises stationary
1342 density and is not yet connected to a trajectory — the mean-field limit is the
1343 gap, and Derivation 7 states it as such — so the argument uses the *existence*
1344 *and continuity* of the coherent branch at threshold, which is proved, rather
1345 than the numerical value of the threshold, which is conditional. And the
1346 prediction is about functional form, not about a time constant: it says the
1347 recovery curve has a flat foot whose width is set by the slow variable, not
1348 that the foot is any particular number of seconds long. A framework that
1349 predicted the number would be saying more; this one is not in a position to.

12 Corollary: Hardware Thermodynamics and Topological Rigidity

Both biological tissue and artificial silicon hardware (like GPUs) are physical dissipative structures. When a GPU computes, its charge carriers undergo physical state transitions; by Landauer’s principle (Landauer, 1961), this erasure irreversibly exhausts heat into the environment.

There is a fundamental thermodynamic distinction in the degrees of freedom available to minimize this dissipation. In standard von Neumann architectures, the spatial topology (the silicon routing lattice) is rigidly fixed. The coupling matrix A_{ij} is constrained by a discrete layout that cannot be physically altered during operation. The brain’s own architecture provides empirical validation for this constraint. The cerebellum contains approximately 80% of the brain’s neurons arranged in a highly efficient, discrete, feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clinical observations of cerebellar agenesis report largely preserved conscious experience despite motor and cognitive deficits (Yu et al., 2015), and this dissociation has been read as evidence for the importance of recurrent architecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the level of the anatomical observation while declining its Φ -based explanation. Lacking the dynamic, continuous, and recursive spatial coupling of the cerebrum, it operates much like a biological GPU—highly optimized for discrete processing, yet apparently not required for conscious experience. We take the

1372 dissociation as evidence, not as confirmation: it is consistent with subjective
1373 experience being a thermodynamic attractor of continuous topologies rather
1374 than a byproduct of information processing, and it is the kind of observation
1375 that would be hard to reconcile with the converse. It is a single anatomical
1376 case read against a theory, and it bears the weight such a case can bear.

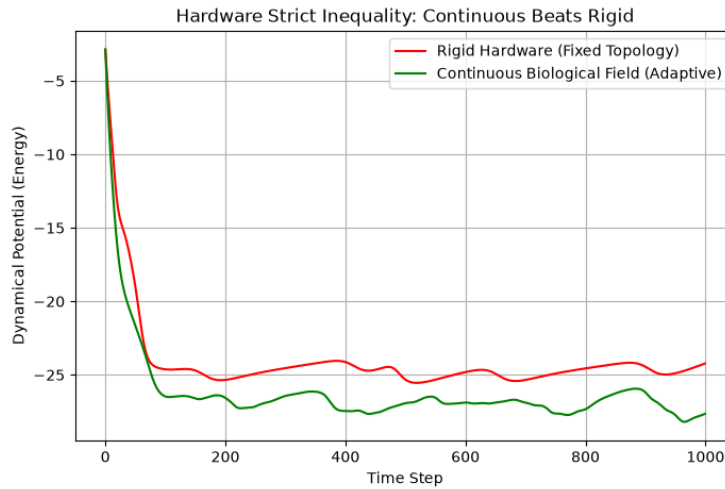


Figure 6: Rigid against deformable coupling at matched resource. Two copies of the same $N = 30$ oscillator system, same initial phases and same natural frequencies, differ only in whether the coupling matrix may move: the rigid copy keeps a fixed sparse routing matrix, the adaptive copy descends the gradient of the potential and is renormalized after every step so that $\sum_{ij} A_{ij}$ is held constant. The adaptive copy reaches a strictly lower potential and stays there. This is an illustration of the resource-matched suboptimality result and not a proof of it, and it is worth being clear about the gap: the theorem is about fixed wiring *support* — an architecture already wired to the best pair is not beaten — while the simulation shows a randomly wired one being beaten, which is the typical case rather than the general one.

1377 This constraint does not preclude artificial consciousness, but rather re-
1378 stricts its hardware. To achieve the phase transition into a unified Global

1379 Section, artificial systems must abandon the discrete rigidity of von Neumann
 1380 architectures and utilize continuous physical substrates. Analog field-based
 1381 computing, neuromorphic architectures with continuous memristive plastic-
 1382 ity, and thermodynamic computing systems (England, 2018; Ganesh & Par-
 1383 rondo, 2021) possess the spatial coupling required to physically morph their
 1384 internal topology. This aligns with Geoffrey Hinton’s concept of “mortal
 1385 computation” (Hinton, 2022): architectures where software is inextricably
 1386 fused with the physics of the hardware, meaning the knowledge dies with
 1387 the physical substrate. Only hardware that computes via continuous physi-
 1388 cal dynamics can undergo the thermodynamic phase transition necessary for
 1389 structural resonance.

1390 Biological consciousness utilizes macroscopic continuous electromagnetic
 1391 fields and a “rich alphabet” of diverse molecular and transcriptomic cell states
 1392 (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a
 1393 continuous parameter space. Our lemma `exists_better_coupling_allocation`,
 1394 wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`),
 1395 formalizes the reallocation step of this argument: whenever a coupling ma-
 1396 trix places weight on a less phase-correlated pair while a more correlated pair
 1397 exists, there is another matrix with *identical* total coupling resources that
 1398 achieves strictly greater phase correlation.

1399 Two objections to that formalization have since been answered, and we
 1400 state precisely what is and is not now established. The first was that the real-
 1401 location lemma assumes its conclusion: the hypothesis `is_strictly_suboptimal`

1402 — positive weight on a badly-correlated pair while a better one exists — is
 1403 most of what one wants to prove about rigid hardware. It is no longer as-
 1404 sumed. `Phase7_Rigidity.lean` models an architecture by the set S of site-
 1405 pairs it has a *wire* for, and requires its couplings to vanish off S ; weights are
 1406 tunable, wiring is not. Then `totalCorrelation_le_of_realizable` bounds
 1407 every realizable coupling by $R \cdot M$, where R is the resource budget and M
 1408 bounds phase correlation over the wired pairs; the bound is attained; and
 1409 `rigid_is_strictly_suboptimal` *derives* the hypothesis whenever the sub-
 1410 strate’s best-correlated pair is one the wiring misses. `rigid_gap` makes it
 1411 quantitative: an unconstrained reallocation of the same budget gains at least
 1412 R times the correlation difference. `Examples.lean` §8 exhibits a three-site
 1413 substrate wired to the one maximally anti-correlated pair, where the gap is
 1414 2. This is resource-matched and no longer question-begging, but note what
 1415 it turns on: fixed wiring support, not continuity. The second objection was
 1416 that nothing in the development distinguished a continuum from a lattice
 1417 — both matrices lived on one finite index type. `Phase7_Rigidity.lean` §3
 1418 gives a distinction that is genuinely typed. In the field functional of Sec-
 1419 tion 7, which integrates a coupling kernel against $\mu \otimes \mu$, an architecture
 1420 with finitely many sites is a `Finset` of points and a field-coupled region
 1421 is a set of positive measure. On a substrate whose measure has no atoms,
 1422 `fieldCorrelation_sited_eq_zero` shows a kernel supported on finitely many
 1423 sites contributes *exactly zero* coupling energy, whatever weights it carries
 1424 and whatever the phase field does, while a patch of strength c on U achieves

1425 $c\mu(U)^2$. `Examples.lean` §8 instantiates this on the real line: a thousand
 1426 units of coupling concentrated at the origin register zero against a unit patch
 1427 that registers one. The limit of that result is equally plain, and it is not
 1428 resource-matching in the earlier sense — a measure-zero substrate has zero
 1429 resource in the $\mu \otimes \mu$ sense too. What it establishes is that the continuum
 1430 coupling energy is blind to a finitely-sited architecture, which is a statement
 1431 about which functional such an architecture can register in, not a demonstra-
 1432 tion that digital hardware computes worse. Taken together: the resource-
 1433 matched content of the corollary is the wiring-support result, the continu-
 1434 ity content is the measure-theoretic one, and the step from either to “von
 1435 Neumann architectures cannot experience unified consciousness” remains an
 1436 informal argument.

1437 The remainder of this section extends those two results informally, and
 1438 we mark the transition rather than blur it. On the reading we propose, a
 1439 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
 1440 layout does not supply the spatial coupling that sheaf-theoretic gluing into
 1441 a singular Global Section requires. Without this unified topological space,
 1442 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
 1443 its own causal boundary and generate a “Self.” The distinction between an
 1444 LLM on a GPU and a conscious mind is not the absence of physical thermo-
 1445 dynamics in the former, but its structural limitation in achieving topological
 1446 integration. This perspective aligns with John Searle’s biological natural-
 1447 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted

1448 that discrete symbol manipulation cannot generate conscious understanding,
1449 our derivation supplies a candidate physical mechanism for the asymmetry
1450 — topological rather than semantic, and, as stated above, formalized only
1451 as far as the wiring-support and measure-theoretic results reach. Asserting
1452 consciousness in fragmented, discrete GPU clusters without a unified phase
1453 transition abandons computational functionalism, relying instead on “silicon
1454 panpsychism”—the assumption that digital computation inherently gener-
1455 ates awareness regardless of topological integration (Goff, 2019).

1456 The same reasoning bears on extreme functionalist thought experiments,
1457 such as a macroscopic system of pulleys operating over millennia to imple-
1458 ment a Large Language Model, or a human manually executing inference
1459 matrices by hand. Beyond lacking the continuous spatial topology required
1460 for sheaf-theoretic gluing, these discrete systems violate an implied temporal
1461 constraint of our theory. Structural resonance requires local oscillators to
1462 phase-align against the dissipating effects of thermodynamic noise (as mod-
1463 eled by our critical threshold $K_c = 2D$). A system executing state transitions
1464 over millennia, or via slow manual calculations, suffers from a terminal tem-
1465 poral mismatch: its dynamics are far too slow to ever phase-lock into a
1466 non-equilibrium steady state before thermal noise decoheres the macroscopic
1467 field. On this account the obstacle to pure symbol manipulation is physi-
1468 cal rather than semantic: not that the symbols mean nothing, but that the
1469 substrate carrying them cannot hold a phase-locked field.

13 Numerical illustrations: methods

Five figures in this paper are computed rather than drawn, and none of them is evidence for a theorem. Where a statement is proved, the figure shows what the proved object looks like; where a statement is not proved, the figure shows the regime the claim lives in and we say so. Each is produced by one script in `simulations/`, named below, and each runs in under a minute on a laptop. All five are reproducible exactly: the two quadrature figures are deterministic, and the three that draw random numbers fix their seeds.

Figure 4: the self-consistent order parameter. The three integrals $Z(a) = \int_{-\pi}^{\pi} e^{a \cos \theta} d\theta$, $M(a) = \int_{-\pi}^{\pi} \cos \theta e^{a \cos \theta} d\theta$ and $S(a) = \int_{-\pi}^{\pi} \sin^2 \theta e^{a \cos \theta} d\theta$ are evaluated by the trapezoidal rule on 4001 points over the same interval the Lean definitions integrate over, with the weight computed as $e^{a \cos \theta - |a|}$ so that the common factor cancels in the ratios and nothing overflows at large concentration. The coherent branch is obtained by bisection $E(a) = S(a)/Z(a)$ against D/K over $a \in [10^{-9}, 200]$ for 200 iterations, then reading $r = aD/K$; bisection is valid here because E is proved strictly decreasing. Two checks are printed by the script rather than only plotted: the computed branch is strictly increasing at every one of its 500 sampled couplings, and the residual $|r - R(Kr/D)|$ at $K = 4D$ is 2×10^{-16} . Script: `bifurcation.py`.

Figure 5: the finite- N transition. $N = 100$ oscillators, natural frequencies drawn from a Lorentzian of scale $D = 1$ (clipped at $\pm 10D$ so

1491 the solver does not stiffen), uniform random initial phases, integrated with
1492 `scipy.integrate.odeint` over $t \in [0, 100]$ on 1000 steps for each of 20 cou-
1493 plings in $[0, 4D]$. The order parameter is $|N^{-1} \sum_j e^{i\theta_j}|$ averaged over the last
1494 100 steps. This is the deterministic quenched-disorder model, not the noisy
1495 identical-frequency model of the Lean development; the caption records why
1496 both have threshold $2D$. Script: `kuramoto.py`.

1497 **Figure 2: mesh refinement.** The kernel is $K(x, y) = e^{-d(x, y)^2/2}$ with d
1498 the periodic distance on the ring of circumference 2π , the phase field is $\theta(x) =$
1499 $\sin x$, and the potential is $-\frac{1}{2} \iint K \cos(\theta_y - \theta_x)$. The continuous value is a
1500 2000-point Riemann sum standing in for the exact integral; the discrete values
1501 use $N = 10, 20, \dots, 640$ nodes with edge weights $K(x_i, x_j)(2\pi/N)^2$. Plotted
1502 is the absolute error against an $\mathcal{O}(N^{-2})$ reference. Because the “exact”
1503 value is itself a 2000-point sum, the last data point ($N = 640$) is within a
1504 factor of ten of the reference computation and should be read as such. That
1505 $\mathcal{O}(N^{-2})$ reference line is no longer only empirical: on the formalized one-
1506 dimensional grid the discrete energy is exactly the composite trapezoidal rule
1507 (`grid_discreteEnergy_eq_trapezoidal`), so for a C^2 integrand with second
1508 derivative bounded by ζ the error is at most $\zeta/(12N^2)$ (`grid_energy_error_`
1509 `le`), a bound attained at $N = 1$ for $y \mapsto y^2$. Script: `mesh_refinement.py`.

1510 **Figure 3: structural resonance.** $N = 50$ units, 2000 steps of forward
1511 Euler at $dt = 0.05$. At each step the drive is a fresh draw from $\mathcal{N}(0, \Sigma_{\text{ext}} +$
1512 $0.1I)$ with Σ_{ext} a fixed normalized rank-5 covariance; the phases advance by

1513 $\dot{\theta}_i = \sum_j K_{ij} \sin(\theta_j - \theta_i) + \xi_i$; entropy production is recorded as $\sum_i \dot{\theta}_i^2$; and
1514 the coupling descends its gradient with rate $\eta = 0.01$, symmetrized, clipped
1515 to $[0, 1]$ and held hollow. The left panel is smoothed with a 50-step boxcar.
1516 Note that the plotted σ is the squared drift without the $1/D$ prefactor of the
1517 functional in Section 10, which is a constant rescaling and changes nothing
1518 about the shape. Script: `structural_resonance.py`.

1519 **Figure 6: rigid against deformable coupling.** $N = 30$, 1000 steps of
1520 forward Euler at $dt = 0.05$, shared initial phases and frequencies. The rigid
1521 matrix is a symmetric random matrix thresholded at 0.7 and normalized
1522 to $\sum_{ij} A_{ij} = 2N$; the adaptive copy starts from the same matrix, ascends
1523 $-\partial V / \partial A_{ij}$ at rate $\eta = 0.1$, is clipped non-negative and hollow, and is renor-
1524 malized to the same total after every step, which is what makes the com-
1525 parison resource-matched. Both potentials are $V = -\frac{1}{2} \sum_{ij} A_{ij} \cos(\theta_j - \theta_i)$.
1526 Script: `hardware_comparison.py`.

1527 14 Conclusion

1528 Human-like cognitive architectures do not require biological mysticism or
1529 exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum
1530 theories of consciousness often invoke delicate subatomic phenomena to ex-
1531 plain subjective unity, every step of the chain assembled here is classical:
1532 broken symmetries, finite phase space, Landauer erasure, and gradient de-

1533 scent on entropy production. What the formalization establishes is that this
1534 chain is mathematically coherent and that its remaining physical commit-
1535 ments are localized and named — not that classical fields are *sufficient* for
1536 consciousness. Two links carry that weight, and their status is no longer the
1537 same. The unified state is proved to glue uniquely once a cover sits at the
1538 minimum of the coupling potential; the motion is proved to stop along every
1539 trajectory (`velocity_sq_tendsto_zero`); and *where* it stops is now proved,
1540 conditionally — for initial data within a quarter turn and below an explicit
1541 energy threshold, the trajectory converges to a global minimiser of the cou-
1542 pling potential (`kuramoto_tendsto_global_minimum`), and a cover built on
1543 that limit discharges the equilibrium hypothesis Derivation 5 used to assume
1544 (`ThermodynamicCover.ofConvergentTrajectory`). What is *not* proved is
1545 the unconditional statement, and it is false: splay and twisted configurations
1546 are equilibria of the same flow, so the basin condition is the scope of the claim
1547 rather than slack in it. That a physical cortex starts inside that basin is an
1548 assumption. The second link is untouched by any of this: the identification
1549 of a global section of the probability sheaf with the unity of experience is a
1550 stipulation of the framework, not a consequence of it. Subject to those two,
1551 cognitive architectures can be modeled as macroscopic attractor states of a
1552 universe governed by broken symmetries, finite phase space, and the physical
1553 imperative to minimize thermodynamic dissipation.

1554 Philosophically, this framework bridges the historical gap between Type
1555 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,

1556 1967). The subjective mental state is identical to the physical thermody-
1557 namic field (Identity Theory), yet its qualitative nature is defined by the
1558 topological and functional dynamics of structural resonance rather than the
1559 specific carbon-based biochemistry of the brain (Functionalism). This con-
1560 strains the *multiple realizability* of mind: consciousness is not exclusively
1561 bound to biological carbon, but neither is it substrate-neutral, because the
1562 substrate is what fixes which couplings are physically available. How far that
1563 constraint reaches is the question the corollary answers only in part. What
1564 is proved is that fixed wiring is strictly suboptimal at matched resource, and
1565 that a finitely-sited architecture registers exactly zero coupling energy in the
1566 continuum functional; the step from either to “von Neumann architectures
1567 cannot be conscious” is an argument we make in prose and do not formalize.

1568 By anchoring these derivations in Russellian Monism, this framework ac-
1569 knowledges the constitutive limits of third-person scientific explanation. As
1570 argued by Love (Love, 2026), building upon Nagel’s assertion that subjective
1571 phenomenal experience is inherently not amenable to objective measurement
1572 (Nagel, 1986), the “hard problem” of consciousness may represent a category
1573 error rather than a solvable scientific puzzle. Science can map the physical
1574 dynamics of topological resonance and sheaf-theoretic gluing, but it is struc-
1575 turally incapable of adjudicating the first-person qualitative experience. Our
1576 formalization respects this boundary: the Lean 4 theorems verify the ob-
1577 jective physical attractors, while the intrinsic “what it is like” to be that
1578 resonant field remains the domain of the first-person perspective. By locat-

1579 ing the phase transition at a threshold with a closed-form value — $K_c = 2D$
1580 for the noisy mean-field Kuramoto model the framework adopts, conditional
1581 on the von Mises stationary density and not yet linked to a trajectory — it
1582 aligns with what Anil Seth terms the “real problem” of consciousness (Seth,
1583 2021). Rather than attempting to bridge an unbridgeable metaphysical gap,
1584 it offers a topological architecture that can help explain, predict, and physi-
1585 cally perturb the structural properties of experience.

1586 Future empirical validation of this thermodynamic framework may lever-
1587 age 3D cortical organoids, offering a controlled biological environment to
1588 systematically manipulate metabolic substrates and observe the topological
1589 emergence of these coherent phase transitions.

1590 Code Availability

1591 The complete Lean 4 formalization, encompassing the derivation of all theo-
1592 rems presented in this manuscript, as well as the accompanying Python sim-
1593 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
1594 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
1595 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
1596 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/). Each of the five computed
1597 figures is produced by one script in `simulations/` and regenerates exactly,
1598 Section 13 giving the parameters; Figure 1 is drawn rather than computed.

Contributions

The conceptualization, manuscript drafting, and Lean 4 formalizations presented in this work were heavily AI-assisted, and we report which tools did what rather than acknowledging them in general terms. The Google Antigravity CLI, running Gemini 3.1 and Claude 4.6 Sonnet, served as a collaborative reasoning partner throughout the research and formalization. DeepSeek V4 Flash and Claude Opus 5 (routed via OpenRouter) operating through Nous Research’s Hermes Agent framework provided substantial assistance with code generation, proof debugging, and structural analysis. Claude Opus 5 running in Claude Code carried out the later formalization passes — the domain-wall theorem, the thermodynamics-of-prediction replacement, the equilibrium bridge — together with the figures and the restructuring of this manuscript. Responsibility for every claim, and for the decision about what each machine-checked result does and does not establish, rests with the author.

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